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1999

SPECIAL FEATURE
The health care system

LUXEMBOURG

OECD ECONOMIC SURVEYS

1998-1999

LUXEMBOURG

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 4 December 1998.

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The previous Survey of Belgium and Luxembourg was issued in April 1997.

BASIC STATISTICS OF LUXEMBOURG (1997)

THE LAND

Area (km^2)	2 586
Agricultural area, 1996 (km^2)	1 264
Woodland, 1994 (km^2)	886

Major city, thousand inhabitants:	
Luxembourg, 1.3.1991	75.8

THE PEOPLE

Population (thousands), 1.1.1997	418.3
Inhabitants per km^2	162
Net natural increase, 1.1.1997	1 794
Net migration, 1.1.1997	3 672

Employment (thousands):	
Total domestic employment	226.5
Dependent employees:	209.9
Agriculture	1.6
Industry and construction	54.6
Services	153.7
Employers, self-employed persons and domestic help	16.6

PRODUCTION

Gross domestic product (billion LF)	587
Gross domestic product per head (US$)	39 244
Gross fixed investment:	
Per cent of GDP	22.9
Per head (US$)	8 992

Gross domestic product by origin, at market prices (per cent):	
Agriculture	0.8
Industry et energy	15.7
Construction	6.5
Other	77.0

THE GOVERNMENT

Per cent of GDP:	
General government consumption	12.4
Central government current revenue	31.6
General government gross debt	6.4

Composition of the Chamber (number of seats):	
Christian-socials	21
Socialists	17
Democrats	12
Others	10
Total	60

Last election: 12.6.1994

THE CURRENCY

Monetary unit: Luxembourg franc

Currency units per US$, average of daily figures:	
Year 1997	35.7584
November 1998	34.7042

Note: An international comparison of certain basic statistics is given in an annex table.

LUXEMBOURG

Assessment and recommendations

Growth has remained robust...

The Luxembourg economy has continued to perform well, with GDP growth picking up to around 4¾ per cent in 1997, and probably also in 1998. Exports have been a major driving force, underpinned by buoyant export markets and a strong export performance of the financial and communication sectors. Private consumption has strengthened due to higher wage increases and robust employment growth. Investment has also been buoyant, partly due to special factors such as the purchase of some aircraft.

... with strong job creation, declining unemployment...

Despite 15 years of strong job creation, tensions in the labour market have largely been absent due to the growing presence of foreign workers. Unemployment has been declining in 1998, and although remaining around a historically high level for Luxembourg, it is one of the lowest in the OECD area. The employment rate is relatively low as many older workers – by moving to early-retirement or disability schemes – have left the labour force. Hence, a broader concept of labour under-utilisation (''broad unemployment''), which adds people of working-age in benefit schemes and labour market programmes to registered unemployment, shows a less favourable development, with broad unemployment exceeding 13 per cent of the broad labour force in 1997 – compared with less than 12 per cent in 1995.

... and virtually no inflationary pressures

Inflationary pressures have been largely absent, and consumer price inflation has followed developments in neighbouring countries. Wage rises edged up to nearly 3 per cent in 1997, but this was mainly due to the institutional wage indexation which resulted in a 2.5 per cent

general wage increase at the beginning of the year. In general, recently-concluded collective labour agreements provide little or no signs of strong wage rises, with contractual real wage increases merely picking up to slightly over 1 per cent. As a result, compensation per employee is expected to decelerate to 1.6 per cent in 1998. Given moderate wage developments, unit labour costs in the manufacturing sector have developed favourably compared with those of Luxembourg's trading partners. But international price competitiveness has been virtually unchanged, as export prices typically move in tandem with those of foreign producers.

The outlook is favourable

Real GDP is projected to grow at around 3½ per cent in both 1999 and 2000. A further increase in the number of cross-border workers should prevent the economy from overheating but will also keep unemployment at around current levels, despite robust employment growth and the implementation of a national employment plan. Consumer price inflation is projected to stay low – in line with developments in neighbouring countries. The risk and uncertainties concern primarily the external sector and, on balance, are on the downside. Apart from the international financial crisis, a major uncertainty concerns the impact of European Economic and Monetary Union on the financial sector, and the degree to which the loss in revenues from foreign exchange dealings is compensated by increased cross-border activities, facilitated by the single currency. In a more distant future, the possible imposition of an EU-wide withholding tax on savings could reduce the attractiveness of Luxembourg for foreign depositors.

Monetary conditions have remained easy...

Within the Belgium-Luxembourg Economic Union (BLEU), monetary conditions have remained relatively easy over the past year and a half. The contractionary impact of the small increase in short-term rates has been more than outweighed by the expansionary effects of a steady decline in long-term rates even though, due to a possible decrease in inflationary expectations, the decline in real terms may have been smaller and the Luxembourg (and the Belgian) franc, in effective terms, has recently appreciated somewhat. As all the Maastricht criteria had been fulfilled, Luxembourg easily qualified to participate in monetary

union from its inception. Hence, monetary policy arrangements have to be adapted. The Central Bank of Luxembourg – as successor of the Luxembourg Monetary Institute – is actively setting up an operational framework to carry out the policies of the European Central Bank from January 1999. Furthermore, new instruments will be introduced, such as remunerated minimum reserves with the European Central Bank, but these changes are unlikely to greatly affect the position of Luxembourg as a financial centre.

... and fiscal policy has become more expansionary, as taxes have come down...

Fiscal policy has become more expansionary, as the government reduced taxes and increased spending on infrastructure projects and social security. According to government projections, the general government surplus could decline from 2.9 per cent of GDP in 1997 to 1.4 per cent in 1998, despite the strong performance of the economy. Income taxes and business taxes have been reduced, in order to preserve the competitiveness of the Luxembourg economy in an international environment characterised by a widespread reduction in tax rates. The normal tax rate for companies came down to 37.45 per cent in 1998 – compared with 40.3 per cent in 1996.

... and central government and social security spending has been growing rapidly

Central government spending is expected to rise by almost 6 per cent in nominal terms in 1998, mainly because of earlier commitments such as the multi-annual wage agreement for the public sector, and new investment projects on infrastructure and school buildings. Social security spending has also been growing rapidly, due partly to policy decisions but – as in the case of health care – also to rapid cost increases and growing demand. New legislation concerning the disability scheme and the introduction of a long-term nursing care scheme may further add to underlying pressure on expenditure. Social security reform – the implementation of which is largely within the confines of the social partners – has been very slow. To prevent a further expansion of the social security sector, it would be useful if the authorities could explore mechanisms for containing expenditure growth, as is already the case for central government spending. Such moves would need to be underpinned by structural reform of the various schemes.

Longer-term concerns centre on the pension schemes

Longer-term concerns centre on the pension schemes in the public and private sector. Although the Luxembourg pension schemes are in a more favourable position than those of many other countries, pension liabilities are building up. Pension reform plans have mainly focused on curbing expenditure in the public sector scheme by bringing it more in line with the less favourable private sector scheme. Recently, parliament approved the government's plans in this area. However, more reforms seem to be called for, such as a better management of the financial reserves of the pension funds and the creation of a funded complementary pension system.

Favourable tax regimes have come under closer scrutiny

The government has pursued an active policy to enhance the attractiveness of Luxembourg as a site for financial and industrial activity. Tax advantages in comparison with neighbouring countries, strict bank secrecy rules, a liberal regulatory environment, and the rapid implementation of EU directives in Luxembourg law, combined with a favourable geographical location at the heart of Europe and a qualified and multilingual labour force have been central in creating competitive advantages in financial services. Globalisation and technical innovation have enabled companies to exploit differences in tax regimes between countries. Tax competition *per se* is welcome as it focuses on delivering public services in a cost-effective way. Moreover, differences in tax regimes between countries also reflect preferences in public provisions as opposed to private provisions of goods and services. However, concerns have been raised about possible harmful aspects of tax competition. At the end of 1997, the EU council chaired by Luxembourg, agreed to a package of measures to tackle harmful tax competition in order to help to reduce tax distortions in the Single European Market. To counter possible negative effects of tax competition in the financial and other service sectors, the OECD has adopted a set of Guidelines to deal with harmful preferential tax regimes in the OECD area. The Luxembourg authorities, specifically objecting to the Guidelines' reference to the removal of impediments to the access of banking information by the tax authorities, have abstained from approving these Guidelines, and are therefore not bound by them.

Diversification policy should yield to structural reform as EU regulations tighten

Another important aspect of the government's industrial policy is the implementation of incentives to attract new industries as part of the diversification policy, within the framework of EU directives. This policy has been successful in creating jobs in the manufacturing sector – most of which have been taken by cross-border workers – although it may also have distorted the allocation of resources and delayed the adjustment process in the regions affected by the restructuring of the steel industry. The tightening of restrictions on state aid by the European Commission will restrain the scope of the diversification policy in future. Hence, the authorities should focus their policies more on structural reform to enhance competitiveness by removing rigidities in the labour market and further pursue regulatory reform.

Structural reform efforts should be stepped up

Given the favourable economic environment, the need for structural reform has been less apparent than in many other countries, and this may have slowed its pace. The National Action Plan for the Promotion of Employment seems more directed at preventing a further deterioration of labour market performance than at improving it. The proposed measures to upgrade skills should be pursued but may show results only in the medium term. On the other hand, the Plan puts great emphasis on training programmes, subsidised employment programmes, and the promotion of entrepreneurship. Relatively little attention is paid to elements aimed at improving the functioning of the labour market by enhancing incentives to work and by creating more flexibility in the economy in general.

More should be done to reduce labour costs...

A number of recommendations in the 1997 *Survey* still require action. The role of the social minimum wage as an instrument to achieve equity goals should be reassessed with regard to its effects on the labour market. For example, a lower minimum wage could be considered for certain groups, such as for young people above the age of 17 and long-term unemployed, to bring wages better in line with productivity if it turns out that low-skilled or inexperienced workers are being priced out of employment. Indeed, if the level of wages required to price low-skilled workers back into the market were unacceptable, due to equity considerations, wages could be complemented by in-work benefits

– while respecting fiscal constraints and paying attention to avoid excessively high marginal effective tax rates. Alternatively, possible negative effects of the social minimum wage on employment creation could be mitigated by relying more on subsidies for employers. In addition, reductions in social security contributions could be targeted more at low-skilled workers. Finally, given the high replacement rate, incentives to encourage people to actively seek a job or a place in a training scheme could be strengthened further, if not by reducing the generosity of the social security system, at least by intensifying the surveillance on availability for work and job search, and by applying even more strictly benefit sanctions for refusing suitable work or training. According to the Luxembourg government, high net replacement rates do not have a significant effect on the length of unemployment spells. Therefore, it prefers to maintain the generosity of the system and improve placement and counselling services. Moreover, sanctions have been applied more often to those unwilling to take up work. Concerning a more degressive age-dependent social minimum wage, the Luxembourg government refers explicitly to engagements undertaken in other international fora, in particular the Council of Europe, whose social charter allows a maximum differential of 25 per cent between the social minimum wage for young persons and adults.

... relax working hour restrictions, and improve labour participation

The authorities should act more forcefully in relaxing working-time restrictions, especially for part-time workers. Given that part-time work is almost exclusively done by women, such measures may be more effective in stimulating female participation than additional wage subsidies for this group. The government's decision to subsidise collective agreements to reduce working hours and at the same time hire unemployed may prove to be costly and inefficient and should be reconsidered. However, concerning labour market flexibility it should be noted that in specific cases, the application of the strict legislation in this area is often relaxed to suit the requirements of individual firms. In addition, more recently, the government has introduced a law in Parliament, widening the possibilities for derogation of collective agreements and making the strict working time legislation more flexible. With a view to raising

labour force participation, recently introduced tighter controls on access to the disability scheme should be maintained.

Good public health care, but at high costs

The public health sector has been an area where expenditure has risen very rapidly. The Luxembourg health care system provides universal and comprehensive health care, in combination with a private health care provision. Rationing of health services, for instance hospital waiting lists, has been avoided. The health status of the population is high by international comparison. These positive results have been achieved at relatively high costs, even though the average age of the population is lower than in most other countries. Health care indicators point to an over-consumption in the hospital sector and of pharmaceuticals. A major problem is that the health system does not have the possibility to regulate the supply of doctors, as every physician practising in Luxembourg is by law accredited to the Luxembourg health system. Remuneration of health professionals, which is largely based on a fee-for-service system, provides little incentive to minimise costs. In addition, the administration of the public health fund (U*nion des caisses de maladie*, UCM) and health-care providers face a soft budget constraint and investment decisions in the hospital sector are often influenced by socio-political considerations rather than based on a cost-benefit analysis.

Measures to address inefficiencies seem insufficient...

To curb health expenditure growth, the UCM has set up data banks to monitor health expenditure in the ambulatory sector and has started to challenge claims in cases of abuse. Such pressures may have resulted initially in a marked slow-down in treatment, but its effectiveness has faded in the meantime. Moreover, recent rulings by the European Court have undermined these control mechanisms, by giving Luxembourg residents more freedom to buy health care in neighbouring countries. In the hospital sector, the system of financing has been changed through the introduction of prospective budgets. Although the new budget system has provided more transparency, it has hardly been used to increase cost effectiveness in the hospital sector. Moreover, a reduction in the number of hospital beds has proved to be difficult to achieve.

... and more needs to be done

Given expected future expenditure pressures, due to technical progress and population ageing, more changes will probably be required to keep the system affordable. The authorities should strengthen the role of general practitioners, making them gatekeepers of the system. A combination of capitation and fee-for-service systems might help in reducing the number of visits. Efficiency in the hospital sector could be improved by benchmarking hospital services and their costs; using lump-sum payments for certain specific medical acts could help in this respect. In addition, the oversupply of hospital beds should be reduced more swiftly, and co-operation intensified with medical establishments in neighbouring regions. Finally, it would be recommendable to revise the tripartite structure of public health care by reducing the influence of the social partners on public health insurance.

Summing up

Overall, the Luxembourg economy has performed very well over the past 15 years, fuelled by buoyant growth of financial and other business services. Since the beginning of 1997, the manufacturing sector has again started to contribute to growth, underpinned by the start-up of new activities, the completion of the steel industry's restructuring, and the recovery in neighbouring countries. Although monetary and fiscal policies, on balance, are unlikely to provide much restraint over the coming years, the chances of overheating are small as the economy can continue to rely on an influx of cross-border workers. However, some imbalances have become apparent, especially in the labour market where, despite strong employment growth, labour force participation has remained low. The emphasis should be on increasing labour force participation. Hence, the pace of structural reform should be stepped up, and focused on improving the qualifications of low-skilled workers in order to promote their integration, on increasing incentives to work and on lifting restrictions on working hours, which have proven to have a detrimental effect in terms of jobs and unemployment.

I. Recent trends and prospects

The Luxembourg economy has continued to perform well. Economic activity has strengthened as production in the manufacturing sector has picked up, and the service sector has continued to expand rapidly. After growing by around 4¾ in 1998, real GDP is projected to grow around 3½ per cent both in 1999 and 2000 (Table 1). Employment creation is expected to remain among the highest in the OECD, but as cross-border workers take most of the new jobs, unemployment may hardly change. Despite the long expansion, the economy has not shown signs of overheating, and inflation has remained subdued.

Output growth has been buoyant...

Activity in the manufacturing sector has picked up strongly since the third quarter of 1996, and production increased by around 7 per cent in 1997 and in the first half of 1998, but is likely to have slowed down in the second half of the year. The turnaround has been most noticeable in the steel sector, where demand has been boosted by the recovery of the European economies and the rebuilding of stocks. However, order books have started to weaken since April, pointing to slower, production growth in the second half of 1998. The improvement in the steel market coincided with the replacement of the last blast furnace by an electrical one in May 1997, which completed a LF 24 billion conversion programme in the steel industry which had started in 1993. Production in the other manufacturing sectors also progressed strongly, partly because of non-recurrent factors such as the expansion of a wood processing plant. Furthermore, production in the construction sector has strengthened significantly since the first quarter of 1998, partly due to a substantial increase in public investment.

The service sector has continued its remarkable expansion and output growth was 6 per cent in 1997 (in current prices), only slightly lower than a year earlier. The financial sector remains one of the main pillars of the Luxembourg economy, contributing 17 per cent of GDP (national version)[1] and 9 per cent of employment. However, growing competition in international banking has led to rationalisation in the banking sector and interest margins have come down

Table 1. **Demand and output: recent trends and projections**

Annual percentage change, 1990 prices

	1995 current prices		1997[1]	1998[1]	1999[2]	2000[2]
	LF billion	Per cent of GDP				
A. Demand and output						
Private consumption	278.8	53.9	2.5	2.6	2.4	2.5
Government consumption	67.2	13.0	1.7	2.8	2.8	2.8
Gross fixed investment	111.4	21.6	14.1	6.5	4.7	5.0
Final domestic demand	457.3	88.5	5.5	3.8	3.1	3.3
Stockbuilding[3]	11.7	2.3	–0.6	0.2	0.2	0.2
Total domestic demand	469.1	90.8	4.8	4.0	3.3	3.5
Exports of goods and services	467.5	90.5	6.0	7.5	5.0	5.3
Imports of goods and services	419.8	81.2	6.1	7.0	5.1	5.5
Foreign balance[3]	47.7	9.2	0.5	1.2	0.4	0.4
GDP at constant prices	..	..	4.8	4.7	3.4	3.5
GDP price deflator	..	..	2.4	1.5	1.4	1.6
GDP at current prices	516.8	100.0	7.3	6.3	4.9	5.1
B. Memorandum items:						
Private consumption deflator	..	..	1.1	1.1	1.3	1.5
Industrial production	..	..	7.1	5.1	2.5	2.6
Total employment	..	..	3.2	3.0	2.5	2.5
Unemployment rate (per cent)	..	..	3.6	3.1	3.2	3.3

1. Provisional figures.
2. Projections.
3. Contribution to growth of GDP.

Source: OECD Secretariat.

steadily to 0.65 per cent in 1997, compared with 1.2 per cent in 1983.[2] Nevertheless, fee-earning activities have substantially increased, and are now making up about 30 per cent of gross earnings.[3] Although the exposure of the banking sector to the Asian and Russian markets is limited, the sector has reacted prudently to recent international developments by substantially increasing its bad debt provisions. Among other remarkable developments in the service sector is the steady expansion of the satellite network. Currently there are eight satellites in operation and the service can be received by around 70 million European households.

... underpinned by strong export growth...

On the demand side, exports have been the main force behind the current buoyancy of the economy as exports of goods, and in particular steel products, have rebounded sharply. Other manufacturing branches, such as the

chemical industry (plastic, tyres and man-made fibres) and machinery have also performed well. Moreover, the start-up of some new industries, such as wood processing has provided an additional boost. Export growth of services, mainly those of the financial and the communication sectors, has also been buoyant. However, as imports accelerated sharply due to the buoyancy of investment, and in particular the purchase of some aircraft, the contribution of net exports to GDP growth was limited to around ½ percentage point in 1997, though it could have exceeded 1 percentage point in 1998.

... booming investment...

The favourable cyclical environment has played an important role in the current investment boom: business investment outside the steel industry has bounced back sharply, thus more than offsetting the scaling down of investment in the steel industry. Moreover, investment has been sustained by the steady expansion of the satellite network, as each year since 1993 one satellite has been launched.

The improvement in the investment climate is reflected by the pick-up of new investment projects, through the mediation by the Board of Development (*Comité de développement économique*). In 1997, 10 new projects were pledged, for a total amount of LF 8 billion (or 6 per cent of total investment), running over several years. Also, investment projects under the industrial diversification law increased sharply to LF 23.5 billion spread out over several years, and supported by LF 3.7 billion of government subsidies, mainly in the form of capital transfers.[4] Nevertheless, the authorities have found it difficult to attract new investors. The small industrial base of the country is a clear disadvantage, as foreign investors make more and more use of mergers and acquisitions to gain a foothold. Moreover, the lack of available industrial buildings for leasing has been a handicap, as new firms often prefer to limit their initial financial commitment.

Residential investment accelerated by 3 per cent in 1997, sustained by low mortgage interest rates and a favourable economic climate. Given the growth in mortgage lending and building permission, the expansion in this sector is likely to have strengthened in 1998. In addition, government investment has grown substantially, owing to an increase in the school building programme and the commencement of the construction of two motorways in 1998.

... and buoyant household demand

Household disposable income has considerably strengthened over the past two years. At their biennial adjustment in January 1997, the social minimum

wage and related benefits were raised by 3.2 per cent. Moreover, in the following month all wages and benefits were adjusted by 2.5 per cent, in conformity with the institutionalised wage indexation (*échelle mobile*). Disposable income was further boosted by a LF 7 billion (1.2 per cent of GDP) income tax reduction in the 1998 Budget. At the same time, national employment has been growing steadily by around 1¼ per cent, and real wage settlements by around 1 per cent. As a result, private consumption seems to have been picking up rapidly and is expected to have grown by 2½ to 2¾ per cent in 1997 and 1998. Government consumption strengthened, also as Luxembourg assumed the Presidency of the European Union during the second half of 1997.

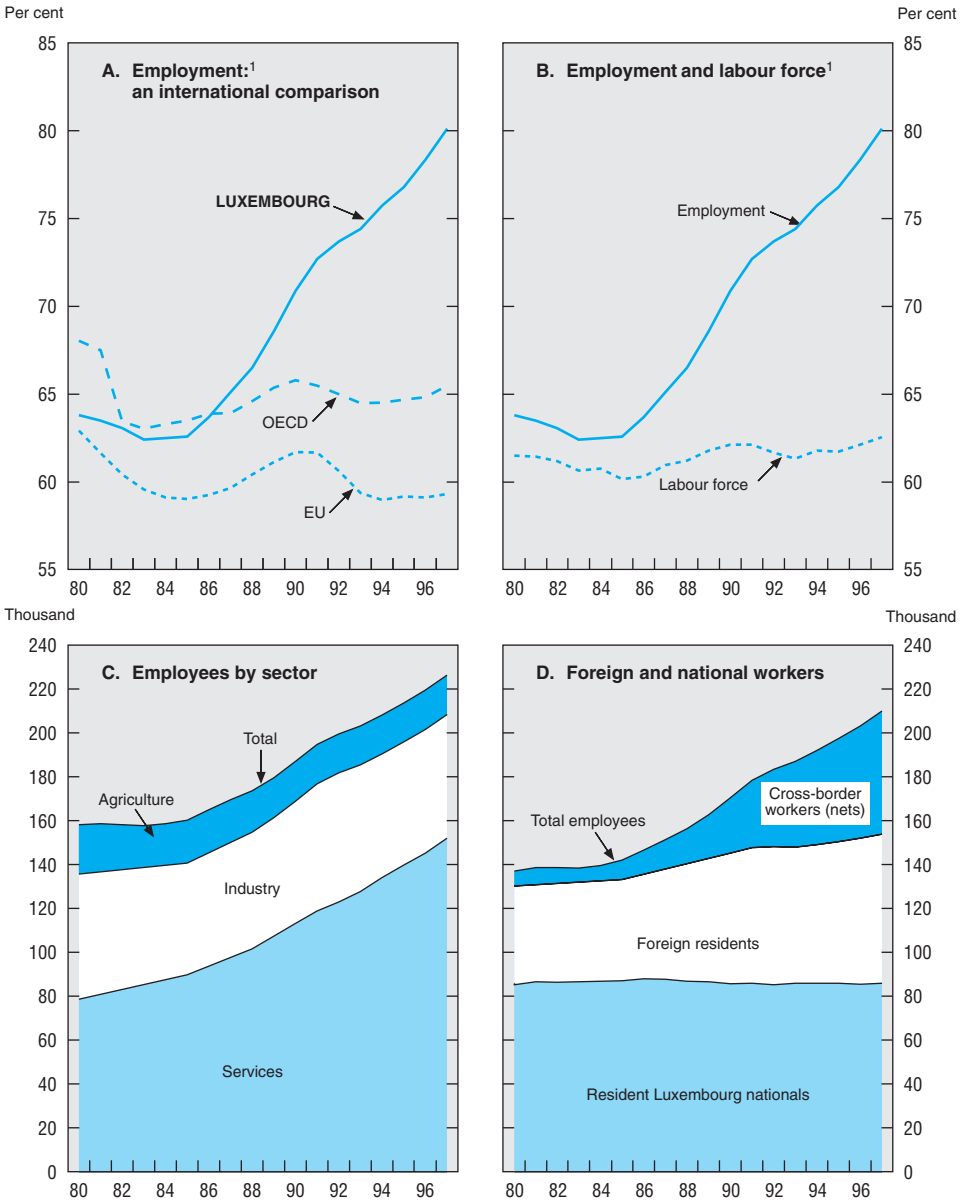
Unemployment has been declining

Total employment growth has remained at around 3 per cent. In 1997, employment in the steel sector declined further by almost 500 persons (8 per cent of the work force in the sector) following the closure of the last blast furnace. However, these losses were more than compensated by further employment creation in the service sector, thus underlining the widening gap between a buoyant service industry and – at least in employment terms – a sluggish manufacturing sector, notwithstanding a substantial increase in manufacturing employment, outside the steel industry (Figure 1, Panel C).

Despite fifteen years of strong job creation, tensions in the labour market have largely been avoided due to a growing presence of foreign workers. The availability of cross-border workers has been stimulated by the relatively high net wages in Luxembourg[5] and the large supply of labour from the surrounding regions. Currently, around 75 per cent of the net increase in jobs is taken by these cross-border workers, whose share in domestic employment has steadily increased to over 30 per cent. Moreover, more than 30 per cent of the population are foreign nationals, and their share in employment is around 25 per cent. Thus the majority of the active labour force does not have Luxembourg nationality (Figure 1, Panel D).

Unemployment has been declining since the beginning of 1998 and, at just below 3 per cent in August, it was 0.5 percentage point lower than a year earlier. But this decline has partly reflected a change in definition, which has entailed the removal of all participants in labour market policy measures from the national unemployment definition.[6] Taken together, the number of registered unemployed and participants in labour market programmes was almost unchanged in August 1998, compared with a year earlier. A broader concept of labour market under-utilisation ("broad unemployment") which adds working-age people in all kinds of benefit schemes and labour market programmes to unemployment, shows that labour market conditions are less favourable than

Figure 1. **Overview of the labour market**



1. As a per cent of the working age population.
Source: STATEC, *Statistical yearbooks* and OECD Secretariat.

suggested by traditional unemployment measures (Table 2). Broad unemployment rose to more than 13 per cent of the broad labour force in 1997 – compared with 11.7 per cent in 1995.[7]

Table 2. **Unemployment, special programmes and broad unemployment**

Annual averages

	1980-85	1990	1995	1996	1997[1]
I. Unemployment	2 077	2 060	5 130	5 680	6 354
As a per cent of the national labour force	1.4	1.3	3.0	3.3	3.6
II. Other unemployed					
Part-time unemployed	126	42	39	113	203
III. Special labour market programmes	2 412	627	1 082	1 396	2 421
Job creation in the steel sector (DAC)	2 216	101	171	144	1 499
Youth work scheme (DAT)	114	380	558	762	714
Training	82	146	353	490	208
IV. Other benefit programmes	10 621	14 244	15 791	16 483	16 888
Early retirement[2]	1 362	2 378	1 421	1 352	1 400
Disability[3, 4]	9 259	11 866	14 370	15 131	15 488
V. Broad unemployment (I + II + III + IV)	15 235	16 973	22 042	23 672	25 866
As a per cent of the broad labour force	9.2	9.5	11.7	12.3	13.2

1. Provisional or estimated data.
2. *Préretraite ajustement* or for shift workers.
3. End of year.
4. Disabled resident in Luxembourg. Data for 1980-90 have been estimated by the OECD Secretariat.

Source: Statec and Inspection générale de la Sécurité sociale.

This less favourable view of the labour market is supported by other indicators, such as the labour force participation rate, which has been low, especially for older employees, and at 61 per cent is one of the lowest in the OECD. One of the reasons for this is that many older workers took early retirement as part of the restructuring of the steel industry. The participation rate for older men (over 50 years of age) has come down to only 30 per cent, the lowest rate in the European Union after Belgium. Another reason is the traditional low employment rate of women. The government has been preparing measures to promote the participation of women in the labour market as part of the National Action Plan for the Promotion of Employment (see Chapter III).

Unemployment still compares favourably with other OECD countries. The contrast is even more striking compared with the surrounding regions: in 1997, the number of unemployed in Luxembourg was only 6 400 (3.6 per cent of the labour

force) compared with almost 200 000 (above 9 per cent of the labour force) in the surrounding regions (Statec, 1998).[8] According to the authorities, the severe unemployment problem in the surrounding regions is one of the reasons for the absence of any improvement in domestic labour market conditions. Other important factors are not only the lack of qualifications – 60 per cent of registered unemployed in Luxembourg have less than nine years of education – but also insufficient incentives to look for work (OECD, 1997*a*). However, since the last *Survey*, counselling and placement services have been improved, and sanctions for those unwilling to take a job have been imposed more often. A particular problem is the growing number of older unemployed (older than 50 years), of which more than a third has been registered with the Public Employment Service for more than a year. Youth unemployment, although almost twice the overall unemployment rate, does not seem to pose a serious problem as only around 10 per cent are registered for longer periods of time.

Moderate wages and low inflation

Wage rises have accelerated to close to 3 per cent in 1997, mainly due to the above-mentioned biennial revision of the social minimum wage and institutionalised wage indexation (Table 3). Taking these two factors into account, there

Table 3. **Compensation per employee**

Percentage changes

	1991	1992	1993	1994	1995	1996[1]	1997[1]
Compensation per employee	6.4	5.3	5.0	4.0	2.2	1.8	2.7
Of which:							
Mining and manufacturing	7.8	4.1	7.7	5.2	6.8	0.6	2.8
Construction	17.9	–3.8	–2.9	1.5	2.6	–0.3	2.8
Market services	5.7	7.1	5.2	4.4	0.3	2.0	2.4
Banking and insurance	7.9	10.2	9.9	11.6	–1.1	2.8	3.0
Non-market services	3.0	7.1	5.7	2.3	3.0	2.4	3.4
Employer's contribution	–0.4	–0.1	0.2	–0.5	–1.5	0.4	. .
Gross wage	6.7	5.4	4.8	4.5	3.7	1.4	. .
Wage indexation (*échelle mobile*)	2.9	3.2	1.8	3.0	1.9	1.0	2.3
Real wage (including wage drift)	3.8	2.2	3.0	1.5	1.8	0.4	. .
Memorandum item:							
Minimum wage[2]	9.6	5.0	7.1	3.5	2.1-7.3	0.8	5.6

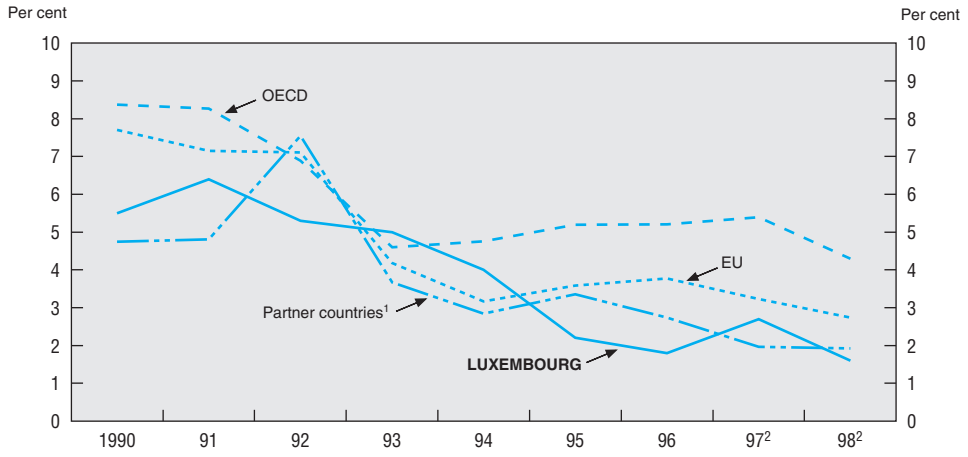
1. Estimates.
2. The distinction between workers with and without dependent family members was abolished in 1995.

Source: Statec, Inspection générale de la Sécurité sociale and OECD Secretariat.

are only faint signs that wage inflation will pick up: new collective agreements in the banking sector and steel industry provide for annual real pay rises of around 1 per cent for 1998 and 1999. Moreover, the growing presence of cross-border workers in Luxembourg has avoided tensions in the labour market and limited upward pressures on wages, as their wages are substantially lower than those of residents.[9] Hence, compensation per employee is projected to decelerate to 1.6 per cent in 1998. Overall, wage developments in the 1990s do not seem to be very different from those in neighbouring countries, despite generally tighter domestic labour market conditions – even allowing for broad unemployment (Figure 2).

Figure 2. **Compensation per employee: an international comparison**

Growth rate, per cent

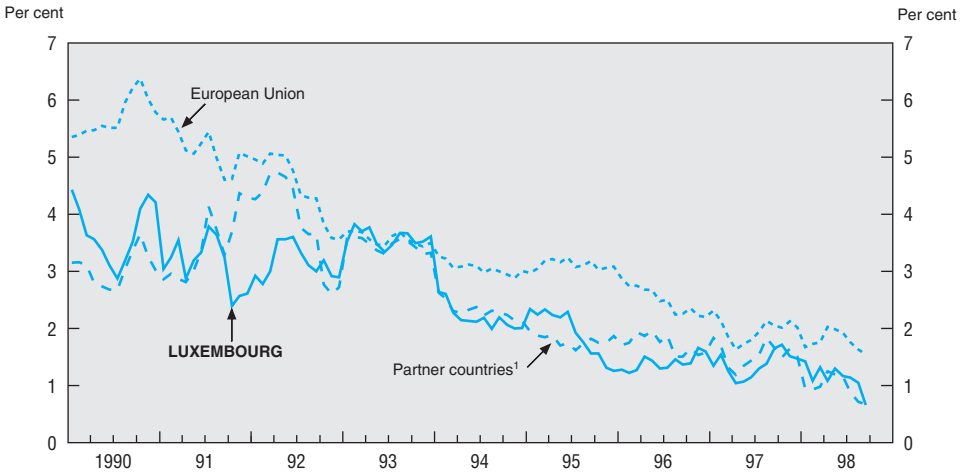


1. Belgium, France and Germany.
2. Estimations.

Source: STATEC and OECD Secretariat.

Consumer price inflation has followed more or less the declining trend in neighbouring countries (Figure 3). After rising for most of 1997, inflation has been coming down, reaching 0.5 per cent in October. In particular, energy prices have contributed to a slowing of inflation, being around 5 per cent lower than a year earlier. These price movements have not been significantly divergent from those in the surrounding countries.

Figure 3. **Consumer prices**
Per cent change over previous year



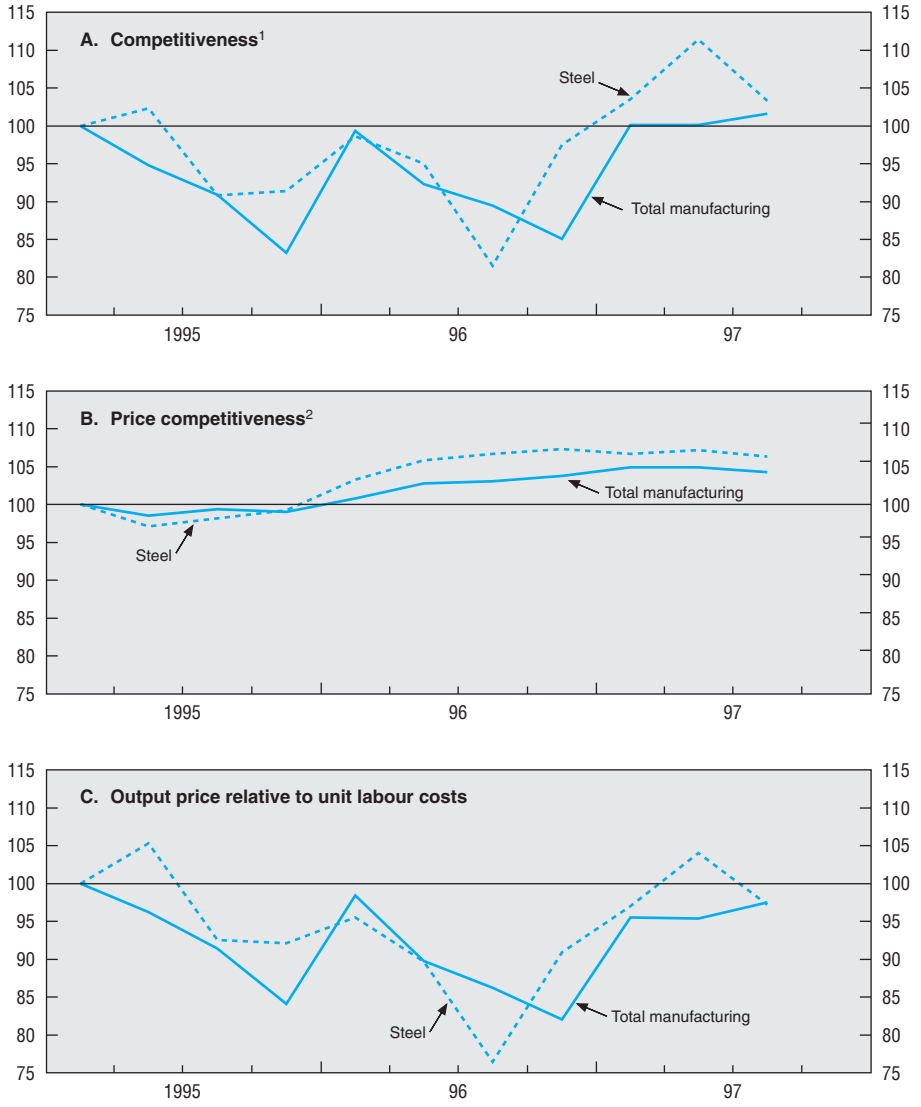
1. Belgium, France and Germany.
Source: OECD, *Main Economic Indicators.*

High current-account surplus

After a sharp improvement during the second half of 1996, international competitiveness in the manufacturing sector – as measured by a competitiveness indicator based on unit labour costs in a common currency – has remained fairly stable (Figure 4, Panel A).[10] Only the indicator for the steel sector saw a further improvement in the first half of 1997 but fell back in the second half of that year. Producers do not seem to have used their improved position to reduce prices as output prices have moved almost in parallel to those of foreign producers (Figure 4, Panel B), as Luxembourg's manufacturing sector is to a large extent characterised by "price taking", particularly in the steel industry.[11] Although unit labour costs have lagged behind producer prices (Figure 4, Panel C), profit margins have not necessarily been increased. Prices for other inputs, such as scrap iron, have risen sharply because of the depreciation of the Luxembourg franc *vis-à-vis* the dollar, and increased demand for raw materials by the steel sector.

Exports of goods (in value) were 10.5 per cent higher in 1997 than a year earlier, largely attributable to a pick up in steel shipments and higher prices for (long) steel products, which cover around 25 per cent of exports. The switch-over to electric steel-making processes, although hardly affecting the overall trade balance, has changed the composition of trade: electricity imports have replaced

Figure 4. **Competitiveness of the manufacturing sector**
Index 1995 Q1 = 100



1. Foreign output prices (in common currency) relative to Luxembourg unit labour costs, (see Krecké and Pierretti [1997]). An improvement in competitiveness is indicated by an increase in the index.
2. Foreign output prices (in common currency)/Luxembourg output price

Source: STATEC.

coke and oil imports since the closure of the last blast furnace, and imports of scrap iron have replaced imports of iron ore as raw materials in the industry. But the trade balance deteriorated sharply, mainly because of exceptional factors such as the purchase of two aircraft in the second half of 1997. The deterioration in the trade balance was only partly offset by an improvement in the service balance, of which more than 50 per cent is generated by the financial sector. Also the exports of communication services strongly increased, owing to the expansion of the satellite network. However, the surplus on the income account deteriorated due to the growing number of cross-border workers. Overall, the current account surplus fell to around LF 70 billion (13 per cent of GDP) (Table 4).[12]

Table 4. **Current balance of payments**

LF billion

	Average 1990-93	1994	1995	1996	1997[1]	1997 S 1[1]	1998 S 1[1]
Current balance of payments	95.0	90.8	84.8	87.0	74.1	33.9	32.3
As a per cent of GDP	23.7	18.2	16.3	15.9	12.6		
Goods and services							
Goods	–55.7	–58.7	–46.9	–51.9	–66.3	–32.4	–29.0
Exports	209.1	216.6	272.4	259.9	308.5	143.1	167.0
Imports	264.8	275.3	319.3	311.8	374.8	175.5	196.0
Services	39.7	46.1	69.2	76.6	82.8	46.3	42.0
Exports	95.7	121.7	246.1	283.5	334.9	160.3	183.1
Imports	56.0	75.6	176.9	206.9	252.1	114.0	141.1
Income (net)	119.3	115.4	80.9	77.7	74.5	29.3	27.3
Compensation of employees	–22.4	–37.5	–43.1	–48.8	–56.0	–26.3	–30.2
Investment income[2]	141.7	152.9	124.0	126.5	130.5	55.6	57.5
Current transfers (net)	–8.3	–12.0	–18.4	–15.4	–16.9	–9.3	–8.0
General government	–4.5	–8.0	–3.9	–0.8	–1.5	–2.1	–2.0
Other sectors	–3.8	–4.0	–14.5	–14.6	–15.4	–7.2	–6.0

1. Provisional figures.
2. Without reinvested earnings.
Source: Statec.

Exports of goods strengthened further in the first half of 1998, and were 20 per cent higher in value terms than a year earlier. As imports were 13.5 per cent higher, the deficit on the trade balance improved only slightly. The current-account surplus remained almost unchanged for the same period, as an improvement in investment income was offset by a further increase in compensation of cross-border workers.

Favourable short-term outlook

Short-term prospects for the economy remain favourable, with the service industry being the main driving force. Already one of Europe's investment fund leaders, Luxembourg looks well positioned to take advantage of the creation of a single European pension market. The government has already prepared legislation to be implemented, once EU legislation has cleared the way for cross-border pension products. In addition, the communications sector is expected to gain further momentum, following the steady expansion of the satellite network. However, growth in the banking sector is likely to slow as a result of the introduction of EMU, as foreign exchange earnings – which are currently 4 per cent of the banking sector gross revenues – will be reduced. The manufacturing sector is projected to slow due to the deceleration of activity in Europe, but also as some of the current dynamism is due to exceptional factors such as the start-up of new activities and the pick-up in steel production after the termination of a substantial investment programme.

The employment situation is unlikely to change much, despite the implementation of a LF 2.6 billion employment plan (see Chapter III). Although employment growth should remain robust, unemployment is expected to stay around current levels, as the high influx of cross-border workers is projected to continue, thus preventing skill shortages and mitigating wage claims. The economy is unlikely to overheat and consumer price inflation could remain in line with that in neighbouring countries.

As far as the risk are concerned, a small economy as Luxembourg depends largely on developments in neighbouring countries, especially for the manufacturing sector, and the current international financial crises could have an important negative impact. In addition, the service sector, and more specifically the banking and insurance industry, depends heavily on European Union legislation, and the removal of non-tariff barriers in the European market, in particular for cross-border pensions. Apart from the international financial crisis, a major uncertainty for the projections is the impact of European Economic and Monetary Union. As mentioned earlier, the banking sector is likely to lose part of its foreign exchange earnings, but the impact may be limited as large part of foreign exchange dealings involves currencies outside the euro area. Moreover, the sector is likely to benefit from increased cross-border activities, facilitated by the introduction of a single European currency. In a more distant future, the possible imposition of a Europe-wide withholding tax on savings, as proposed by the European Commission, could reduce the attractiveness of Luxembourg for foreign depositors and might have negative consequences for the Luxembourg banking industry.

II. Economic policies

The Luxembourg authorities have been operating in a favourable economic environment: general government finances have been in surplus for over a decade and the debt-to-GDP ratio – at around 7 per cent – has been well below the criterion of the Maastricht Treaty. On the monetary side, long-term interest rates have continued to trend down, reaching historically low levels, at least in nominal terms. On the fiscal side, policy has become more expansionary, as government investment has increased sharply, in particular for schools and motorways, and taxes have come down. Future challenges centre on the affordability of the social security system, and in particular the pension system. Furthermore, the process of globablisation has increased the need for international co-operation on taxation, and tax incentives have come under closer scrutiny by the European Union and the OECD.

Monetary policy

Owing to economic union with Belgium, Luxembourg does not have an independent monetary policy and interest rates are closely linked to Belgian rates. Hence, as in Belgium, money market rates have edged up while long-term rates have continued to trend down, reaching historically low levels, at least in nominal terms. Due to the convergence of long-term interest rates with German rates, Luxembourg franc issues dropped sharply in 1997 as investors favoured other EU currencies with higher interest rates, such as the Italian lira or the Spanish peseta. Also currencies outside the prospective monetary union area were in favour, such as the Danish krone or currencies with potential exchange-rate gains such as the US or Australian dollar. The market for Luxembourg franc issues, as well as for issues in all other national currencies of the euro area, will disappear with the start of European Economic and Monetary Union from the beginning of 1999.

The Luxembourg economy fulfils all the Maastricht criteria and the European Commission decided in the middle of 1997 that the economy had achieved sufficient convergence to join monetary union. Membership has led to a number

of changes in domestic monetary policy arrangements and the functioning of the Luxembourg Monetary Institute (IML). The IML became the Central Bank of Luxembourg (*Banque centrale du Luxembourg*) on joining the European System of Central Banks (ESCB) in June 1998. The primary objective of the Central Bank is to maintain domestic price stability, while its basic tasks are to implement the monetary policy of the European Central Bank; to conduct foreign exchange operations; to hold and manage the official reserves of Luxembourg; to promote the smooth operation of payment systems; and to issue currency and manage its circulation. The Bank shall continue the supervision of that part of the financial sector for which it has competence (banks, investment funds, other financial institutions, except insurance undertakings). In accordance with the Maastricht Treaty, the Bank is independent insofar as ESCB-related tasks are concerned. Currently, the government is preparing a bill to split competences: the Central Bank of Luxembourg will carry out central bank activities in the strict sense; and the *Commission de surveillance du secteur financier* the prudential supervisory tasks.

The new central bank is setting up the infrastructure to carry out monetary operations. The European Central Bank has announced that remunerated minimum reserves will become one of the monetary instruments. As minimum reserves were not imposed in the past in Luxembourg, the authorities initially feared that their introduction might weaken competitiveness and could result in a relocation of financial activities. But, as the reserves will be remunerated by the prevailing repo rate, the possible negative effects have been minimised. Other monetary instruments will be standing facilities and open market operations.

Fiscal policy

Overview

Luxembourg's fiscal performance compares favourably with that of other OECD countries, with the general government account having been in surplus for over a decade. Generally, the social security sector – in particular the pension scheme – has been in surplus, and the accounts for central government and local authorities have been in surplus or close to balance. Although no formal medium-term target for the general government balance is set, a budget norm exist for central government and balance requirements for the different social security schemes. Central government spending is subject to an expenditure norm, based on a medium-term real GDP growth rate and the institutionalised wage indexation mechanism (*échelle mobile*). In practice, the norm is applied in a flexible way to allow for exceptional expenditures. Moreover, due to the lack of historical data, the medium-term growth rate is based on estimated GDP growth in recent years, and varies from year to year: in the 1997 budget proposal, the medium-term

growth rate was set at 3 per cent, while in the 1998 budget proposal it was set at 3.6 per cent. In addition, large investment projects are not subject to the budget norm and are financed outside the central government budget by special investment funds, though government funding of these funds is subject to the norm. Hence, in recent years, current expenditure has exceeded the budget norm without capital spending being squeezed.[13] According to the authorities, the part of central government spending that is subject to the budget norm has edged down, from around 31 per cent of GDP in the early 1990s to around 29 per cent in 1998. However, once special funds are taken into account, central government spending has slightly increased from 29 per cent in 1994 to 30 per cent in 1998 (Table 5).[14]

Balance requirements exist in the social security sector, where the different schemes are legally required to maintain minimum reserves. However, social security spending is not subject to a norm, and has steadily increased as

Table 5. **Central government budget**[1]

LF billion

	1994 Final	1995 Final	1996 Final	1997 Provisional	1998 Estimate
Primary receipts	148.7	156.0	171.1	185.2	186.7
as a per cent of GDP	29.8	30.1	31.3	31.6	30.1
Of which:					
Indirect taxes	62.0	65.2	72.7	77.0	77.3
Direct taxes	75.0	77.7	85.7	92.1	93.5
Primary expenditure	144.0	156.1	164.3	176.3	186.7
as a per cent of GDP	28.9	30.1	30.1	30.0	30.1
Of which:					
Consumption	42.0	47.8	49.4	54.2	54.4
Transfers to other sectors	29.0	24.3	26.1	28.3	29.2
Transfers to local government and social security	49.7	59.2	63.1	67.4	69.2
Investment and capital transfers	21.7	23.1	23.5	23.9	31.0
Primary surplus	4.7	–0.1	6.8	8.9	0.1
Net interest payments	–0.7	–0.6	–1.0	–1.4	–1.8
Financial balance	5.4	0.5	7.8	10.3	1.9
as a per cent of GDP	1.1	0.1	1.4	1.8	0.3
Memorandum items:					
Budget growth norm	4.7	4.9	3.8	5.9	5.1
Expenditure growth (excluding special funds)	8.7	3.6	11.0	2.2	0.9

1. National definition. This table is not comparable with Table 6 which is on an SNA basis. The accounts of special state funds have been consolidated with the central government account.

Source: Inspection générale des finances and OECD Secretariat.

a per cent of GDP, sometimes due to policy decisions but – as in the case of health care – also due to rapid cost increases and growing demand in the existing schemes (see also Chapter IV). If surpluses in the social sector fall short of their legal requirements, statutory benefits have to be reduced, contributions have to be raised, or central government has to increase transfers to the social security sector. It should be noted that such transfers are subject to the budget norm of the central government.

Recent budgetary outcomes for 1997 and the 1998 budget

According to provisional estimates, the 1997 budget outcome was substantially better than projected and the surplus in the general government account edged up to 2.9 per cent of GDP in 1997 (Table 6). Tax revenues were expected to have slowed, following tax reductions for enterprises. However, as a result of tax windfalls, in particular in the financial sector, tax receipts were 6 per cent higher than a year earlier. Government investment surged owing to the start of some large school projects. The general government debt was – at only 6.4 per cent of GDP – very low compared with other OECD countries.

Table 6. **General government budget and debt**

SNA basis, LF billion

	1994	1995	1996[1]	1997[1]	1998[2]
General government net lending	13.6	9.5	15.3	17.1	8.4
As a per cent of GDP	2.7	1.8	2.8	2.9	1.4
Central government	4.8	–0.1	6.9	8.9	0.1
Local government	1.0	1.7	2.7	1.9	0.1
Social security	7.8	7.9	5.7	6.3	8.2
General government gross debt	27.6	30.0	34.7	37.8	42.3
As a per cent of GDP	5.5	5.8	6.3	6.4	6.8

1. Provisional.
2. Estimates.
Source: Inspection générale des finances and OECD Secretariat.

Assuming medium-term GDP growth at 3.6 per cent, the government fixed the budget norm for 1998 at 4.25 per cent. According to the government, almost 90 per cent of the budgetary expansion was already earmarked for increases in salaries and pensions for civil servants and an increase in child benefits by LF 1 000 for each child per month. The budgetary impact of the latter was offset by a similar reduction in the dependency exemption for each child in the income

tax, as part of the restructuring of child support. This change will be especially beneficial for low-income families, who will profit little from the reduction in income tax, as also proposed in the budget. Hardly any room was left for new spending initiatives within the central government budget, and expenditure restraint was budgeted to fall on government employment and capital expenditure. However, many of the new spending initiatives, especially for infrastructure, would be financed outside the budgetary framework by special funds. Thus, total government investment was projected to increase by more than 25 per cent – reaching more than 3 per cent of GDP – as a start was made with the construction of two new motorways.

In response to intensified international tax competition, the government continued to reduce company taxes, to match more favourable tax regimes elsewhere. The corporate tax rate was again reduced in 1998 to 30 per cent and – under certain conditions – the wealth tax became deductible from the corporation tax. Moreover, the system of tax provision in the financial sector was broadened and enterprises were allowed to make special provisions for the costs associated with the introduction of the euro in 1999. As a result, the normal tax rate for companies came down to 37.45 per cent in 1998 compared with 40.3 per cent in 1996. The budget also contained measures to reduce income taxes by LF 7 billion (1.2 per cent of GDP), partly to neutralise the effects of inflation on tax brackets. Despite these measures, tax receipts were again projected to increase. However, total central government receipts may have fallen as transfers from local authorities dropped sharply, due to changes in the financing of nursery and primary education. On this basis, the surplus of the central government budget may have almost completely disappeared. In addition, the surplus in the local authority sector is also expected to have been lower. However, the overall surplus of the social security sector may have improved, owing to a temporary rise in the contributions for health insurance to bolster the reserves of the health fund to the legally required level. Overall the authorities expect the general government finances to have remained in surplus in 1998. The public sector debt is expected to have increased to 6.8 per cent of GDP, as the government has taken over the debts of the national railway company.

The 1999 budget and beyond

In the 1999 budget, central government expenditure is projected to rise by 5.9 per cent, well below the 1999 expenditure growth norm (6.6 per cent). Central in the budget is the National Action Plan to Promote Employment, with total costs of LF 2.6 billion (0.4 per cent of GDP), mainly financed by the Employment Fund (see Chapter III). The Plan foresees the introduction of a parental leave scheme (LF 1.5 billion), half of which is financed by the budget. The government also participates for 45 per cent in the financing of a long-term

nursing scheme (initial total costs LF 6.2 billion), which will be introduced in 1999. As in 1998, child benefits will be increased by LF 1 000 for each child, as part of the restructuring of child support. Government investment remains at a relatively high level (3 per cent of GDP), due to the continuing construction of two motorways and some school building projects. Furthermore, the government's multi-annual investment plan includes the construction of a new light railway, the so-called "B*us*-T*ram*-B*ahn*". The government expects that the central government budget (excluding the special funds) will show only a slight surplus.

The current fiscal position of Luxembourg is healthy and budgetary problems are not expected in the near future. However, a potential problem is the rapid expansion of government expenditure, especially in the social security sector. The introduction of a long-term nursing care scheme and the widening of the disability scheme may further add to underlying pressures on expenditure. Currently population-ageing problems are less serious in Luxembourg than in most other countries, because of the relatively young population and the heavy reliance on cross-border workers, who currently contribute more to the system than they receive in benefits. However, these workers are building up substantial claims on the social security system and the number of pensions has already been growing faster than the number of contributors to the pension scheme. Nevertheless the reserves of the general pension scheme are still beyond the legal limits.

At the end of 1995, the government commissioned an actuarial study of all existing pension schemes in Luxembourg. The study showed that – with zero employment growth in the public sector – there will be two civil servants for each retired civil servant by the year 2030, compared with a ratio of three civil servants per retiree in 1994. Following this report, the government prepared legislation for a new pension bill in the public sector, bringing it more in line with the less favourable scheme in the private sector. It entails the continuation of the current system for retired employees; a transitional phase for current civil servants; and the setting up of a scheme for future civil servants, comparable to the one in the private sector. Public sector pensions will be reduced from 83.3 per cent of final income to a minimum of 72 per cent. Even under the new system, public sector workers still wanting to receive their so-called 5/6 pension can do so by working until they are 65. The pension reform bill was recently adopted by parliament.

Further measures may be necessary to preserve the long-term sustainability of the pension system. The financial position of the pension fund could be further improved by a better management of the fund's reserves. Currently, the reserves are exclusively invested in Luxembourg, partly in real estate and other long-term assets and partly in short-term assets. The tripartite Social Economic Council has recommended that social funds' management of reserves should be improved, by broadening the range of investment products and giving

the funds the legal possibility to invest in stocks. Furthermore, the authorities could consider to create a funded complementary pension system for the private and public sector.

Tax policies to promote financial and industrial activities

The government has pursued an active policy to enhance the attractiveness of Luxembourg as a site for financial and industrial activity. The absence of monetary reserve requirements for banks, tax advantages over neighbouring countries,[15] a liberal regulatory environment, and the rapid implementation of EU directives in Luxembourg law, combined with a favourable geographical location at the heart of Europe and a qualified and multilingual labour force have been central in creating competitive advantages in financial services. Luxembourg's traditional insistence on protecting the confidentiality of banking information, (except in clearly defined circumstances, such as money laundering) has also contributed to its success as a centre for financial activity.[16] Today, more than 200 banks have been established in the country, accounting for 17 per cent of GDP (national version, *cf.* footnote 1), 20 per cent of central government revenues and 9 per cent of employment.

Recent years have brought increasing attention to the international aspects of taxation and of the need for tax co-ordination, particularly within the European Union. Various factors have contributed to this interest such as the process of globalisation and the development of global strategies by multinational enterprises. Technological innovation has affected the way in which multinational enterprises are managed, making a physical location of management and other service activities much less important. International financial markets continue to expand. The OECD has long recognised that fair competition in this new environment encourages government to deliver public services in a cost-effective way. Moreover, differences in tax levels and structures between countries reflect preferences in public versus private provisions of goods and services. However, these developments, as noted in the recent OECD Report on Harmful Tax Competition (OECD, 1998*a*), opened up the danger that governments may develop tax schemes as well as non-tax measures to attract geographically highly mobile activities. Globablisation also opens up new ways for individuals and companies to minimise their tax liabilities by exploiting differences in tax systems between countries. Thus the spread of preferential tax schemes may encourage activities, which are primarily aimed at promoting profit-shifting without corresponding shifts in real activities. This form of tax competition is regarded as harmful as it carries the risk of distorting trade and investment and could lead to the erosion of national tax bases.[17]

To counter the negative effects of tax competition on international welfare, the EU and the OECD recently issued reports on this subject. At the end of 1997, the EU council, chaired by Luxembourg, agreed to a package of measures to

tackle harmful tax competition in order to help to reduce tax distortions in the Single European Market. The package includes a Code of Conduct on business taxation. The OECD has adopted a Report including a set of Guidelines, limited to financial and other service activities (OECD, 1998*a*). Although recognising the need to combat harmful preferential tax regimes, the Luxembourg authorities find the approach of the OECD Report and Recommendations to counteract harmful tax competition partial and unbalanced. In particular, they object to the assessment that bank secrecy is a criterion to identify such regimes. Thus, the authorities have abstained from approving the OECD Report or its Recommendations and are not bound by them.[18]

Another aspect of the government's policy to improve the economic structure is the implementation of financial aid and other incentives to attract new industries as part of the diversification policy. The policy aims at improving the economic and regional structure and is especially targeted at regions confronted with severe job losses due to the restructuring of the steel industry. As EU regulations restrict this type of industrial policy and general investment grants are no longer permitted with the exception of small and medium sized enterprises, a new industrial diversification law came into force in 1993, restricting investment grants to regional aid, and introducing three subsidy schemes for small business investment, R&D, energy-saving and environment-protecting investment. During the period 1976-1995, more than 120 projects profited from these measures and created more than 10 000 new jobs in the business sector. However, the effects on national employment were limited as resident and non-resident foreign workers have taken 75 per cent of these jobs (Statec, 1997*b*). The projects approved in 1997 are expected to create 1 000 new jobs, almost 50 per cent more than a year earlier.

The diversification policy has been successful in creating jobs in the manufacturing sector, although it may have resulted in a sub-optimal allocation of resources by substituting administrative judgement for market judgement as to the most promising industries or growth areas for the economy. The authorities feel that this risk is minimal, as given the size of the economy, they are in much closer contact with industry than authorities in other countries. Another objection to the policy is that these tax incentives (or subsidies) do not seem to address a clearly identified market failure but are mainly motivated by social or political objectives such as maintaining employment in the steel producing regions. In the present context of almost full employment, subsidies may be especially harmful because they may have delayed the adjustment process. The tightening of restrictions on state aid by the European Commission will restrain the scope of the diversification policy in future. Hence, the authorities should focus their policies more on structural reforms to enhance competitiveness by removing rigidities in the labour market and further pursue regulatory reform (see Chapter III).

To preserve international competitiveness, the government has aimed at reducing the tax wedge, and notably employers' social security contributions, by increasing transfers to the social security funds. Currently, social security contributions constitute only 50 per cent of the receipts of the social security sector.[19] In 1996, the total tax wedge (including social security contributions) of the average production worker in Luxembourg was 34.5 per cent, compared with 56.4 per cent in Belgium (OECD, 1997*b*). Recent actions in this area include the successive increases in government transfers to the child benefit system (*allocation familiales*) to replace the employers' contributions to the system in 1994 and to finance the increases in child benefits in 1998 and 1999. The Social Economic Council (*Conseil économique et social*, 1998) has warned against the growing share of government funding of the social security budget, as it may reduce the role of the social partners in this sector. Furthermore the Council expressed concern that this policy might cause imbalances in the long term: with pension liabilities rapidly building up, additional tax receipts – generated by new jobs – could become inadequate to finance the increase in social security spending.

III. Implementing structural reform: a review of progress

Introduction

Within the framework set out by the OECD Jobs Study, the 1997 OECD *Economic Survey of Luxembourg* (OECD, 1997*a*) provided a set of detailed policy recommendations to improve the functioning of the labour market (Box). These recommendations were derived from a review of structural features of the Luxembourg economy, which determine employment possibilities and the level of structural unemployment. The analysis emphasised that Luxembourg's unemployment problems were minor in comparison with those in other European countries, but that unemployment was on a rising trend. Moreover, labour force participation is relatively low, partly reflecting that a large number of people of working age were in other benefit schemes, such as disability and early retirement, and had effectively left the labour market. In addition, female participation has been low, as many women prefer to stay at home to look after their children, which is facilitated by high family incomes, generous child benefits and income tax deductions for children.

The increase in unemployment over the past two decades may to some extent reflect an increasing mismatch between supply of and demand for labour, against a background of a lack of sufficient skills and wage adjustment. Due to the restructuring of the steel industry and automation in the banking sector, job opportunities for low skilled workers have largely disappeared, and the demand for higher skilled and qualified jobs has risen. The authorities also attribute the slow rise in unemployment to the poor labour market conditions in regions within commuting distance from Luxembourg, where unemployment has remained at around 9 per cent, compared with only 3 per cent in Luxembourg. During the past 20 years, an increasing number of people from these areas have found work in Luxembourg, attracted by good job prospects and relatively high (net) wages. Currently, these cross-border workers take around 75 per cent of jobs created in Luxembourg.

Box. Implementing the OECD Jobs Strategy – an overview of progress

Since the previous *Survey*, a number of policy measures in the spirit of the OECD Jobs Strategy have been either implemented or prepared. However progress has been slow and uneven and a few important measures have not been considered.

Job strategy proposal	Action taken	OECD assessment/ recommendations
1. Increase wage and labour cost flexibility		
– Reform the wage bargaining system, phase out wage indexation and consider relaxing the administrative extension of sectoral wage agreements.	The government has submitted a new law on collective agreements to Parliament, giving more possibilities for derogation of collective agreements.	Improve flexibility in wage bargaining and remove wage indexation.
– Reduce the minimum wage and its indexation.	The minimum wage was twice increased in 1997, due to the biannual revision and the statutory wage indexation, and will be increased again in 1999.	Reduce the impact of the minimum wage by either reducing the minimum wage for certain groups, such as young people above the age of 17, in combination with in-work benefits or by subsidies to employers.
– If for political or social reasons wage flexibility cannot be increased sufficiently, further reduce taxes and employers' social security contributions for low-skilled workers.	Social security contributions for blue-collar workers were temporarily increased in 1998.	Reduce social security contributions for low-wage earners or at least for blue-collar workers.
2. Reform unemployment and related benefit schemes		
– Reduce the generosity of the social security system and especially the unemployment benefit scheme.	The generosity of the system has been maintained but placement and counselling services have improved and sanctions have been applied more often.	Assess whether this is sufficient and strengthen incentives to look for work or a place in a training scheme at least by intensifying the surveillance on being available for work and job search, and an even stricter application of sanctions for non-compliance.

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Job strategy proposal	Action taken	OECD assessment/ recommendations
– Reduce the withdrawal rate of benefits of additional earnings in the general assistance scheme (RMG).	No action has been taken in this area.	Improve incentives to look for work for people in the RMG.
– De-couple the housing assistance from the RMG.	Housing assistance rules will be changed in the revised RMG scheme.	Assess whether this has been sufficient.
– Tighten access to early retirement and disability schemes.	Access to some early retirement schemes has been relaxed. A new law has been introduced in parliament broadening the scope of the disability scheme. Medical criteria for disability have been tightened.	Tighten access to the early retirement schemes. Re-examine beneficiaries at regular intervals and promote their integration in the labour market.
3. Expand and enhance active labour market policies		
– Apply more strictly the working requirement in the RMG scheme.	The Public Employment Service has intensified the individual support of the young and long-term unemployed. Benefits will be withdrawn from those who refuse to participate in training programmes. The Public Employment Service will be reinforced and remodelled.	Assess whether recent steps have been sufficient.
4. Increase employment flexibility		
– Relax employment protection legislation and ease restrictions on fixed-term contracts.	The strict working hours legislation for part-time workers will be relaxed on a temporary basis. The government has submitted a new law to Parliament, allowing for a relaxation of the working hours regulations.	Continue to reform the strict working hours legislation and remove obstacles for part-time work and temporary work contracts.

Luxembourg's strict labour market regulations are similar to those in surrounding countries, although the social and economic context of Luxembourg has often allowed pragmatic and flexible solutions. An additional important element of labour market flexibility is provided by the continuous increase in cross-border workers. However, some impediments to (national) employment growth could be identified, such as restrictions on fixed-term contracts or strict employment protection rules, which may have an adverse effect on the hiring of long-term unemployed. Other major impediments stem from a lack of incentives to look for work, reflecting generous social welfare benefits and high marginal effective tax rates. The authorities have taken measures especially to promote employment for the young and long-term unemployed. Among the many specific suggestions, the *Survey* stressed the need to:

- reform the wage bargaining system, abolish wage indexation and reduce minimum wages;
- reduce the generosity of the social security system and, more generally, increase incentives to return to the active labour market;
- apply more strictly the working requirement in the social assistance scheme (*revenu minimum garanti*, RMG);
- relax employment protection legislation and restrictions on fixed-term contracts.

Recently, the government presented a National Action Plan to Promote Employment (*Plan d'action national en faveur de l'emploi*) in connection with EU guidelines, and prepared by a tripartite commission (*Comité de coordination tripartite*) chaired by the Prime Minister. Reflecting the priorities established by the European Council, the Plan is centred around four pillars: the improvement of employability, the development of entrepreneurial skills, the encouragement of adaptability in businesses, and the strengthening of equal opportunity policies. The measures proposed by the National Action Plan are a compromise between government and social partners. Its purpose is not only to combat unemployment but also to preserve existing employment, to increase the participation of the resident population, and particularly of women, and to target young people and long-term and older unemployed or those that risk becoming long-term unemployed. Special attention is given to education, training and lifelong learning programmes. The authorities estimate the costs of the Plan to be around LF 2.6 billion, partly to be financed by an increase of 1 franc on the social security charge on petrol. This increase will not spill over into the institutionalised wage indexation mechanism.

Increase wage and labour cost flexibility

No progress has been made in improving wage flexibility but the government recently submitted a new law on collective agreements to Parliament, which

gives enterprises more possibilities for derogation of collective agreements. The authorities are not in favour of abolishing the institutionalised wage indexation, as it is in their view a guarantee for social peace. The indexation will only be suspended during periods of serious economic crises as laid down in the 1975 wage indexation law.[20] Concerning wage negotiations, the 1994 agreement between the government and the social partners (employers and employees) to moderate wages is still in force, stating that real wage increases should lag behind productivity growth. The biennial revision of the social minimum wage has not been changed and the social minimum wage was increased by 3.2 per cent in January 1997, following the real wage increase in the total economy. Moreover, the Prime-Minister announced during his State of the Nation address that social minimum wages will be adjusted again in January 1999 for past real wage increases, underlining the government's policy that all residents should share in the rising overall prosperity.

Another way to promote employment for low-skilled people would be to lower non-wage labour costs for workers at the lower end of the wage scale. The government has not introduced new measures to reduce non-wage labour costs, given the lack of response from employers to existing programmes for long-term and older unemployed. On the contrary, non-wage labour costs for blue-collar workers were increased by 0.8 percentage point in 1998, in order to restore the reserves of the health funds. This increase is temporary and health insurance contributions are set to come down again in 1999.

Reform unemployment and related benefit schemes

Access to the disability scheme should have been tightened, following several court rulings stating that disability in the current legislation is incompatible with the possibility of exercising a full-time job. However, strong growth of new disability pensions continued unabated at least until recently, as well as the decline in the average age of the disabled. Hence, the authorities have concluded that the actual practice of granting disability pensions is likely to differ considerably from the legal provisions. To end ambiguities, the government is preparing a new disability act. In the new act, a distinction will be made between general disability and occupational disability, similar to legislation in France and Germany. A person will be declared generally disabled if his potential earnings are less than a third of the social minimum wage. If the earning capacity is reduced by more than 50 per cent, a person can receive an occupational disability pension, which is 50 per cent of the general pension benefit but can be cumulated with income from other professional activities up to a certain ceiling.[21] Recently, medical criteria for disability were tightened, resulting in a lower inflow into the disability scheme.

The 1997 *Survey* recommended tightening access to early retirement schemes. The National Action Plan, however, includes some measures to facilitate access to these schemes. The government has announced that it will reduce some constraints, that have limited the use of the part-time early retirement scheme (*préretraite progressive*).[22] Also, in the solidarity early retirement scheme (*préretraite-solidarité*) rules for replacing an older worker by an unemployed person will be relaxed. However, it should be noted that the extensive use of the early retirement schemes is set to end, now that the restructuring of the steel industry is complete. Furthermore, no changes have been made to the unemployment benefit scheme, which was judged in the previous *Survey* to be too generous, in particular in terms of net replacement rates. According to the authorities, the high net replacement rations do not have a significant effect on the length of unemployment spells. Only 20 per cent of unemployed stay for the full period of one year in the unemployment scheme.[23] Therefore, they have preferred to maintain the generosity of the system and have improved placement and counselling services for the unemployed. Moreover, sanctions have been applied more often to those unwilling to take up work.[24] Other problems earlier identified concern the disincentives to look for work in the social assistance scheme (*revenu minimum garanti*, RMG), where the transition to work might leave people in some cases worse off. The government has announced a review of the RMG, including a revision in the calculation of the housing assistance, which will reduce the high marginal effective tax rate for people in this scheme.

Enhance active labour market policy

Active labour market policies for older unemployed have not been very successful. In 1997, only 150 people used wage subsidy programmes for older and long-term unemployed (*aide à l'embauche de chômeurs âgés et de longue durée*), although more than 1 000 registered unemployed would have qualified for these programmes. The lack of skills has been often an obstacle for employers to take on a long-term unemployed. Hence, the, the National Action Plan emphasises the improvement of skills and competences. As part of the Plan, a new training scheme (*stage de réinsertion professionnel*) will offer trainee positions for long-term and older unemployed in enterprises, lasting for up to a year. Participants will receive a benefit equal to the statutory minimum wage, 50 per cent of which will be paid by the Employment Fund (or 65 per cent in the case of women) and will get priority in recruitment if a suitable position in the enterprise arises.

Although youth unemployment does not seem to be a serious problem, the authorities have also decided to strengthen programmes targeted at this group (*Stage de préparation en entreprise*, *Division d'auxiliaires temporaires* (DAT), and *stage-initiation*), by increasing wage subsidies for employers to 50 per cent (or 65 per cent for women). But, the duration of the DAT contracts will be reduced from two

years to one year, to prevent people staying in these programmes for long periods. These programmes could offer young people a chance to enter the labour market, and 70 per cent of participants have found regular employment through them. However, the dead-weight costs of these programmes are likely to be relatively high: given the short unemployment spells of youngsters, many of them would have probably found a job without these programmes.

Increase employment flexibility

Progress in this area – which in Luxembourg falls in the first instance within the purview of the social partners – has been painstakingly slow. Social partners have been able to agree only on the necessity of modernising the current strict legislation concerning working hours.[25] The government has announced the relaxation of rules on part-time work. Working hours will now be calculated on the basis of the minimum reference period of four weeks. However, this measure will only be applied on a temporary basis to verify its impact on the labour market. In addition, the government has submitted a new law on collective agreements to Parliament, including a minimum reference period of four weeks for all workers. The law will also offer the possibility to fix longer or shorter reference periods (with a maximum of a year) by collective agreement. Furthermore the government will examine the possibilities of supporting the transition from full-time to part-time work. This is already the case for older workers in the part-time early retirement scheme. In addition, the government has undertaken to provide financial incentives in the case of collective agreements that seek to reduce working hours and, at the same time, to recruit unemployed workers. If registered unemployed workers are hired, the Employment Fund pays their social security contributions – ranging from 22.4 per cent to 32.1 per cent of the gross salary – for five years. Although it is too early to assess the effectiveness of this scheme, experiences in other countries suggest that the costs of such programmes can be very high, and their effectiveness in terms of aggregate employment limited. Dead-weight losses could be large, if employers and employees would have agreed on reducing working hours, even without the subsidy. In addition, it is not clear what will happen after five years when the wage subsidy expires and the employer is confronted with a substantial increase in labour costs.

The Public Employment Service (ADEM) and temporary employment agencies signed an agreement for closer co-operation from the beginning of 1998. The aim is to make it easier for job seekers registered at ADEM to contact employers using the services of temporary employment agencies. The temporary employment agencies will notify ADEM of work offers, initially lasting a minimum of two weeks; they will also be involved in the administrative monitoring of employment offers. Meanwhile, ADEM will notify these firms of job offers that are clearly identifiable as temporary.

Improving labour force skills and competences

In the framework of the National Action Plan, the apprentice scheme will be reorganised to improve co-operation between schools and enterprises. In addition, the number of apprentice places will be substantially increased, especially for craftsmen. Links between the different courses in the general education and vocational education systems will be improved to facilitate the reorientation of pupils. Furthermore, the number of two-stage apprenticeship courses, that were set up for pupils with learning difficulties, will be enlarged. Finally, the apprentice scheme – which mainly addresses youngsters between 15 and 18 years – will be opened to adults.

The government has reached agreement with the social partners concerning the regulation of lifelong learning schemes and introduced a bill in parliament. A recently opened centre for professional training (C*entre national de la formation professionnelle continue*, CNFPC) will unfold new initiatives for lifelong training programmes. In addition, the government continues its support to schooling activities by branch organisations, and organisations of employers and employees. The government and social partners are also studying new projects to improve the employability of the unemployed, by providing financial support to schemes where the unemployed are temporarily taken on to replace workers that take leave of absence for reasons such as maternity, parental, and sabbatical leave. Although such schemes may be useful in upgrading the skills of the workforce, it is questionable if they should be subsidised. In general, the financing of paid-leave schemes should be settled in the private sector – as is the case with other working time arrangements – and be part of the negotiated wage agreements. Furthermore, the Danish experience with such publicly-financed schemes has been mixed. The generosity of the system encouraged many people to take paid-leave, thereby creating shortages in some segments of the labour market. As a result, the Danish authorities have abolished some of the schemes and scaled back the generosity of others.

Increasing product market competition

The European articles on competition policy, and in particular article 85 and 86, have been implemented in the Luxembourg legislation. Although strict licensing laws may have limited the entry in various professions, which could have hindered the dynamism of certain sectors, the openness of the economy and the proximity of foreign suppliers may have provided a check on abuses of market power. To promote entrepreneurship, the government has embarked on a policy of relaxing existing regulations and reducing administrative burdens. The relaxation of the establishment law in 1997, has allowed enterprises to widen their activities and has resulted in an improvement in their management. The authorities are preparing a relaxation of the strict rules on certificates in certain branches

and abolition of limitations on the number of subsidiaries a person can own. Instead, some minimum certificate requirements will be introduced for managing a subsidiary. Furthermore, the government is intending to relax working-hour regulations for bakeries. In the area of reducing administrative burdens, a pilot scheme is currently underway to streamline procedures for setting up enterprises by introducing a one-counter system for administrative declarations. In addition, the burden of statistical data collection for Intrastat and Statec will be reduced. Finally, attached to every legislative proposal will be an evaluation of its impact on small and medium-sized enterprises (SMEs).

Administrative reform

In 1996, the government approved a major administrative reform programme, which focused on externally and internally oriented reforms. The externally oriented reforms aim at improving the quality of the public service. The government is focusing on a number of practical measures to be implemented in the short and medium term, such as the publication of a users' charter and the improvement of information to the public. The internally oriented reforms aim more at improving the internal organisation of the administration and its managerial methods, in order to improve the quality of the service. The most noteworthy initiatives in this field are the overhaul of public accounting and financial management procedures, the creation of a computerised personnel management system, merging of the *Gendarmerie* and the police force, revision of existing mechanisms for the recruitment into the civil service and improving the integration of trainees in the administration. Furthermore, the government is in the process of opening up the public service to nationals from other EU countries, following a judgement by the European Court of Justice in 1996.

Assessment and scope for further action

As economic growth has been robust and labour market conditions have remained satisfactory, the government has not engaged in a broad structural reform programme. Indeed, the good macroeconomic performance over the past fifteen years does not seem to indicate that the economy is lacking in dynamism. However, despite the current buoyancy, unemployment has only slightly come down. Labour market reform has been slow as decisions in this area are taken on the basis of consensus between the social partners. This consultation process between social partners is an essential element of the Luxembourg model, and has been a guarantee for social peace.

Given that unemployment remained relatively high at the low-wage end of the labour market, the National Action Plan rightly put the emphasis on raising the productivity of low-skilled workers by education, training, and lifelong

learning programmes. However, such programmes will have an effect only in the medium term. Moreover, these programmes will not reach all low-skilled people. Therefore, in the short term, the government should consider reducing non-wage labour costs for the lower skilled or encouraging lower (minimum) wages for certain groups, such as for young people above the age of 17.[26] The reported success of subsidised youth training schemes is a clear indication that the high minimum wage for this group, often lacking in sufficient experience, is a handicap in finding a job. Also evidence from other OECD countries suggests that young people face less employment opportunities if a high minimum wage exists (OECD, 1998*b*). More generally, a lower minimum wage combined with in-work benefits may create new jobs, enhance incentives to work, and keep the standard of living of low-skilled workers at a socially accepted level. In terms of aggregate employment creation, this may be more effective than wage subsidies targeted at specific groups: studies in other countries show that such schemes have high dead-weight and substitution effects as employers may release people who do not get a subsidy and replace them with people who do.[27] However, the effectiveness of an in-work benefit depends on several features of the labour market, the wage distribution and benefit systems. The extra finance needed for the benefits could be quite substantial. If it is to be raised through income taxes, a large group of workers – depending on the wage distribution – may experience an increase in marginal tax rates, thus encouraging them to reduce their hours of work. If the financing needs are reduced through a high claw-back rate, this may lead to very high marginal effective tax rates for certain income ranges. Hence, poverty traps could spread to new segments of the labour market.

In addition, priority should be given to increasing participation in the labour market and in particular by women and older people. In this context, the authorities could proceed more forcefully in taking measures to promote part-time work, by further removing obstacles. Given that most part-time work is done by women, such an approach may be more effective – in terms of costs and net employment gains – than subsidies for women in training schemes, given the high dead-weight and substitution effects usually associated with such programmes. However, for reasons mentioned above, promoting part-time work arrangements through subsidised schemes should be avoided. Furthermore, the authorities should maintain the recently introduced tighter control and access to the disability schemes. Although the proposed legislation in this area broadens the scope of eligibility, the reform is welcome as it allows people in the new professional disability scheme to participate in the labour market. However, the authorities should continue to carefully monitor the growth of the number of beneficiaries in both schemes and consider further measures to limit the schemes to those in genuine need.

IV. The Luxembourg health system

Introduction

The quality of the Luxembourg health service is high and access is universal. Although health care spending, as a percentage of GDP, is modest compared with other OECD countries, it is very high in absolute terms, especially when considering the favourable age structure of the population. While the Luxembourg health system shows similarities with systems operating in Belgium, Germany and France, its specific traits are the cover it provides for a large group of non-residents – essentially cross-border workers – and the use of health services in neighbouring countries. In 1992, some cost containment policies were introduced but they did not result in a deceleration of health expenditure and further reforms seem necessary to contain expenditure pressures due to technological progress and population ageing. The first section focuses on the size of the Luxembourg health system and overall resource use. The second section describes the main features of the system and identifies potential inefficiencies and inappropriate incentives. The third section concludes with some recommendations for further policy initiatives to improve the efficiency of the system.[28]

The size of the health sector

Luxembourg national health spending[29] as percentage of GDP – at 7 per cent – is below the OECD average, but spending per head is one of the highest among the OECD countries (Table 7).[30] The statutory health insurance – to which almost the entire population belongs – finances around two-thirds of national health expenditure. As in many other OECD countries, health expenditure is rising sharply. During the period 1990-1997, nominal health expenditure rose by more than 8 per cent a year on average, more or less in all areas of medical care, raising its share in GDP by 0.5 percentage point.

Traditional explanatory factors for health expenditure, such as the size of the insured population, the number of elderly and price of health care, do not seem to provide an explanation for the rapid increase in spending. Concerning

Table 7. **Health expenditure in relation to GDP and per capita, 1997**

	As a percentage of GDP	Per capita (dollars, using PPP exchange rates)
Luxembourg	**7.1**	**2 340**
Belgium	7.6	1 747
France	9.6	2 051
Germany	10.4	2 339
United Kingdom	6.6	1 347
United States	13.5	4 090

Source: OECD *Health Data 98*; OECD *Purchasing Power Parity Statistics*.

demographic pressures, the Luxembourg health system seems in a better position than those elsewhere since the average age of the insured population is relatively low and falling. One of the reasons for the favourable age structure is the large number of foreign workers (residents as well as cross-border workers). This has two effects: first, their average age is lower than that of the Luxembourg residents and second, in accordance with EU rules, those foreign nationals who return to their country of origin will often be at the charge of their national insurance system.[31] Hence, many older people with high medical costs leave the Luxembourg health system. Medical expenditures have been rising in many OECD countries with very different types of health care systems, suggesting common factors such as technical progress (see, for example, Newhouse, 1992); but in the case of Luxembourg, the particular institutional structures (and associated incentives) for funding and providing health care may also have been conducive to the rapid increase in spending.

In general, the quality of health services has been high. In a recent European survey, about 70 per cent of Luxembourg respondents expressed satisfaction with the health system, which is substantially higher than the average level of satisfaction in the European Union (50 per cent).[32] Moreover, the life expectancy at birth, at 73 for men and 80 years for women, ranks among the highest in the OECD area. In addition, infant mortality – at 4.9 per thousand births – is very low by international standards. However, indicators for resource use point to relatively high inputs, in particular for hospital services. The number of hospital beds – at 10.7 per 1 000 persons – is well above the OECD average and the average hospital admission rates and duration of stay have also been higher than in most other countries.[33] Besides, one has to take into account that Luxembourg residents also use hospitals in neighbouring countries for treatments which are not available in their own country. In addition, the consumption of pharmaceuticals per head – outside the hospital sector – is about 10 per cent above the OECD average.

Problems with the institutional setting

Main features of the system[34]

The Luxembourg social insurance model for the provision of health care shows similarities with systems operating in Belgium, France and Germany. There is only one public scheme, administered by the U*nion de caisse de maladie* (UCM), whose board is made up of representatives of employers and employees, with a single government representative having the casting vote.[35]

Universal coverage is a key aspect of the statutory health system. Affiliation to the scheme is obligatory for anybody who exercises a professional activity in Luxembourg, people receiving an income-replacing benefit, or students if they are not already covered.[36] The system also provides coverage for a large group of non-residents such as cross-border workers and their dependants. This group has grown substantially in recent years and currently accounts for around 20 per cent of those insured.

Solidarity is another major feature of the health-care system. Contributions are linked to taxable income and independent of health risk, and with household dependants co-insured without additional contributions. The insurance aspect of the system has thus been very limited. The contribution rate – set by the UCM – is dependent on expected health expenditures and the legal obligation to maintain reserves between 10 and 20 per cent of actual expenditures. The current contribution rate is 5.1 per cent of taxable income, equally divided between employees and employers.[37] Almost 40 per cent of the public insurance system is funded by the central government in two ways. First, the government pays for certain benefits, such as normal maternity benefits. Second, it pays an extra premium of 250 per cent on contributions from pensioners and 10 per cent on contributions from the active population.

Each medical act which is covered by the insurance is noted down in the so-called nomenclature with its corresponding reimbursement. The nomenclature is jointly decided by the Ministers of Health and Social Security, on the basis of recommendations from the C*ommission de nomenclature*, the medical college and the C*onseil supérieur des professions de la santé*.[38] Acts which are not included in the nomenclature can be reimbursed on advice from the medical inspection of the social security. Given the size of the country, it would not be feasible to offer treatment for all pathologies in Luxembourg. Hence, the insurance reimburses treatment received abroad, although conditional on permission from the medical inspection. Such permission cannot be refused if treatment is not available from a Luxembourg health institute.

Medical treatment is mainly provided on a fee-for-service basis. As health services are almost completely covered by insurance, a meaningful market price cannot be observed. Hence, the UCM concludes conventions with organisations of

health practitioners, which are legally binding for all medical practitioners in Luxembourg. The law sets minimum requirements for these conventions, such as the determination of fees for all medical acts covered by the insurance. The Minister of Social Security can submit the convention to the *Conseil supérieur des assurances sociales*[39] if it contravenes existing laws and regulations. The law also sets a calendar for negotiations, including mediation procedures if the negotiating parties cannot reach agreement. A mediator can be nominated and in case mediation fails, the *conseil supérieur des assurances sociales* can ultimately determine the convention. Medical fees are negotiated individually – mostly without comparison with fees for other medical procedures or similar acts in neighbouring countries – and are adjusted annually according to the average wage increase in the economy, taking into account the increase in the total volume of medical acts. Only in the case of new medical treatments, do fees in neighbouring countries and Germany in particular, serve as a guideline.

Patients have complete freedom of choice of physician and hospital, and rationing of health services, such as hospital waiting lists, has been avoided. To reduce over-consumption, co-payments – up to a certain ceiling – have been introduced for prescription drugs and doctor visits.[40] Co-payments represent about 10 per cent of total medical consumption (excluding non-reimbursable drugs) and the average co-payment for reimbursable prescription drugs is about 20 per cent.

Because of the relatively high reimbursement rate and the comprehensive package of the compulsory health scheme, the role of private health insurance is very limited. The main health insurance company outside the public health scheme is the *Caisse médico-chirurgicale mutualiste* (CMCM). The CMCM insures mainly supplements for medical treatment in hospitals in Luxembourg or abroad. Although 50 per cent of persons in the public scheme are members of the CMCM, the benefits reimbursed by the CMCM amount to only 2.2 per cent of those reimbursed by the public health scheme. However, lack of competition does not seem to have resulted in administrative inefficiencies and may even have some advantages. The UCM has used its monopsonistic powers to obtain information from health providers, and improve the efficiency of health spending through benchmarking, limiting unnecessary care and challenging unjustified claims. Moreover, as a single insurer, the UCM has been able to take a broader view of health care by participating in preventive health care programmes.

Problems and challenges

As in many OECD countries, the major problems facing the Luxembourg health system are the control of expenditure and the reduction in the oversupply of medical services. The authorities have taken several initiatives to slow down health expenditure, mainly related to an improved control of medical

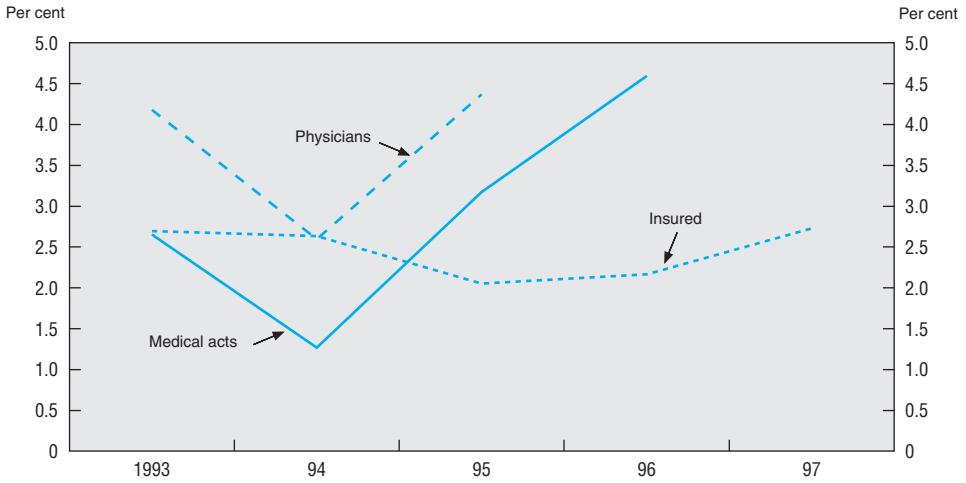
expenditure and pressure on doctors with high prescription records. One of the results of the 1992 reform of health insurance has been the setting up of medical data banks by the UCM, thus enabling them to monitor health expenditure better, and in particular the consumption of reimbursable drugs. Furthermore, the reform altered the financing of the hospital sector by the introduction of a prospective budget system in 1995.

However, control possibilities are limited as a large and still growing proportion of insured persons is not resident in Luxembourg. They are reimbursed for consulting health practitioners in their own countries, which are not bound to Luxembourg health conventions. The control mechanisms of the UCM have been further undermined by the European Court of Justice, which recently ruled that restrictions by the Luxembourg health system on buying health services from non-domestic providers are against the free movement of goods and services in the European Union, and thus illegal.[41] Given the proximity of foreign health practitioners for Luxembourg residents, these court rulings may have serious consequences for the current health system.

Medical care

Although negotiated doctor's fees have risen only moderately, spending on medical care has progressed rapidly as the increase in the number of medical interventions has outpaced the growth of the number of insured (Figure 5). This development may be partly attributed to technical progress, but also the financing method for medical care – mainly a fee-for-service system – and the growing number of medical practitioners may have induced an increase in the number of treatments. The Luxembourg health system does not have the possibility to regulate the supply of doctors, as every physician practising in the country is by law accredited to the health system. Concerning the remuneration system, research in other countries shows that fee-for-service systems tend to lead to a substantial oversupply of health care, and may even go against the best interests of the patient.[42] Patients – not facing the full marginal cost for their decisions – may easily consume more than is strictly necessary and even get reimbursed if they consult several doctors for the same sickness episode. Moreover, lacking sufficient information to make an informed choice, they depend on the decision of medical professionals, who also provide the services required, thus creating a potential conflict of interests. In addition, patients can directly consult a specialist, without prior consultation with a general practitioner. Thus, specialists are in direct competition with general practitioners for patients. Furthermore, the ambulatory sector is not strictly separated from the hospital sector and specialists often continue the treatment of their patients in the hospital sector. Given an oversupply of hospital beds and the lack of incentives to select the most cost-effective treatment for their patients, doctors may too easily opt for hospital treatment instead of cheaper out-patient care. The UCM has tried to curb expenditure by

Figure 5. **Growth of medicals acts**



Source: Inspection générale de la sécurité sociale, UCM.

setting up data banks to monitor medical treatments more closely. This approach was initially quite successful, and the number of interventions decelerated sharply in 1994, when the system was put in place. However, its effectiveness has faded over time. Moreover, it may only have eliminated the most obvious cases of abuse of the system.

Hospital sector

Hospital care has been a major cost component and has risen quickly. As noted earlier, several indicators – such as the number of beds per head – point to substantial over-capacity in this sector. In addition, many of the hospitals are relatively small. Until 1994, hospitals received a flat-rate fee per occupied bed, thus giving them an incentive to prolong length of stay. As this fee was relatively low, hospitals would try to recuperate their costs by overcharging for ancillary hospital services and increase the output of their laboratories to balance the books. Hence, the reimbursement system encouraged unnecessary expenses and distortions in hospital management. Nevertheless, the hospital sector was heavily in deficit by the end of 1994.

The financing of the hospital sector was drastically changed in 1995 by the introduction of a budget system based on prospective spending. Hospitals receive an annual budget to cover their fixed and variable costs. However, all

services by physicians remain on a fee-for-service basis.[43] The budgets are negotiated between the UCM and the 16 hospitals, on an individual basis. Furthermore, the UCM paid off the past debts of the hospital sector, the so-called *Altlasten*.

According to the authorities, the aim of the new system was not primarily to reduce costs, but essentially to improve transparency in hospital funding. Although it may enable the UCM to put more pressure on hospitals with above-average costs to increase efficiency, the current approach provides few incentives for producers to improve the efficiency of their operations, as funding is not contingent on the quantity and quality of output, and little information on relative prices of treatment is generated or used. As negotiated contracts are based on historical costs, the current process even penalises efficient producers and puts little pressure on inefficient providers to improve. Furthermore, budgets are automatically adjusted for wage-cost increases, *i.e.* increases resulting from the wage indexation and collective labour agreements. However, hospital budgets are not automatically adjusted for changes in staffing and skill mix of hospital workers, as they are negotiated between the hospital and the UCM. In fact, in these negotiations trade union representatives are on both sides of the table: they negotiate with the hospital management on the size and composition of hospital staff, while their representatives in the UCM negotiate with the hospital management on the size of the budget. Thus in 1997, labour costs in the hospital sector increased by 6 per cent, compared with 3 per cent in the overall economy.

The authorities have been engaged in restructuring the hospital sector, but closing or merging regional hospitals have proved to be difficult. Currently the 1994 national hospital plan is in the process of being revised. However, the Social Economic Council has already criticised the updated hospital plan as it does not fix objectives nor establishes norms in relation to the country's needs (*Conseil économique et social*, 1998). Big investment projects in the hospital sector (currently above LF 120 million) are for 80 per cent financed by the state. The remaining 20 per cent are paid by the UCM, included in the annual hospital budget. These projects are the responsibility of the Minister of Health, advised by the *Commission permanente pour le secteur hospitalier*, which includes representatives of the Ministry of Social Affairs and the UCM, and the decisions often depend on the presence of the necessary medical expertise in Luxembourg and socio-political considerations. However, financial aspects do not seem to play an important role and the Ministry of Social Affairs and the UCM are only involved in these decisions through their membership of the *Commission hospitalier*. The financing of smaller investment projects are directly negotiated between the UCM and the hospitals.

Pharmaceuticals

The consumption of pharmaceuticals has been relatively high compared with other countries. The authorities have taken several measures to try to

dampen the growth. First, co-payments have been in place for a long time. Prescription drugs have been classified in four categories, ranging from complete reimbursement to zero reimbursement, with the majority of drugs in the 80 per cent reimbursement category. Second, the UCM has set up a data bank, specifying all partly or completely reimbursable medicines outside the hospital sector by patient and prescribing physician and bought in Luxembourg. The UCM has used this data bank to monitor the prescription behaviour of physicians and, if necessary, uses pressure to correct it. In individual cases, the insurance fund may even approach the patient. Despite these measures, the consumption of drugs has increased rapidly, especially for completely reimbursable drugs.

The increase in spending on pharmaceuticals is also due to price increases, well ahead of the rate of consumer price inflation. The authorities fix prices for pharmaceuticals on the basis of producers' (or importers') sales prices in their domestic markets. At the moment, around 75 per cent of pharmaceuticals are imported from Belgium, and the rest from Germany and France where prices are generally higher. However, the share of more expensive German pharmaceuticals is increasing.

Scope for further reform

So far the rapid growth in health expenditure has not posed a serious problem as the economy has been strong and the fiscal position remains favourable. However, given expected future expenditure pressures, due to technical progress and population ageing, more changes will probably be required to keep the system affordable. To reduce moral hazard, the UCM could consider setting limits on the current system in which patients have complete freedom to consult specialists and general practitioners. Although such a system may be advantageous in certain circumstances, there is some evidence that in general, it may be more costly than a system where the general practitioner acts as a gatekeeper, *i.e.* patients need a referral from the general practitioner to get access to specialist care or hospital care (see Gerdtham *et al.*, 1994). In conjunction with this, the remuneration system could be changed. The general practitioner could, for instance, be remunerated by a combination of capitation payment with a small fee-for-service component.[44] The UCM could monitor the system to avoid cream-skimming by doctors, trying to off-load costly patients.

The UCM could also consider an increase in co-payments. However, the effectiveness of such a measure partly depends on the price elasticity of health services, which is probably low. Moreover, the measure is not effective once a patient has reached the ceiling on co-payments or in cases where employers reimburse the co-payments to their employees. Employers have only limited means to reduce health spending by their employees (and their dependants) and an increase in co-payments may only result in higher labour costs.

The UCM has made a start by introducing prospective budgets in the hospital sector. However, these measures have hardly been used to encourage greater efficiency and effectiveness by hospitals. Nevertheless, the current budget system has been an improvement by giving the health fund more insight into hospital spending. As a next step, the health fund could start benchmarking hospitals and introduce lump-sum payments based on the Diagnostic Related Groups-based system.[45] Similar systems have already been implemented or considered by other OECD countries.

The over-supply of hospital beds is a serious problem. Decisions to close hospitals are always politically sensitive, especially in a small country such as Luxembourg. A tighter budgetary system may go some way to forcing hospitals to co-operate, and may result in the merger of some smaller hospitals. Moreover, important savings could be made if co-operation between hospitals were not only restricted to Luxembourg but also involved those in neighbouring regions, the so-called G*rande région transfrontalière*. As a starting point, the Ministry of Social Affairs and the UCM need to be more closely involved in major investment decisions in the sector. In such a way, cost arguments could be incorporated in decisions on introducing new technologies in Luxembourg or buying therapies from hospitals in neighbouring countries.

Finally, many arrangements in the health sector are determined between the health fund (UCM), health practitioners, trade unions and employers' organisations. In some cases, the same negotiating parties are even on both sides of the table, such as in negotiations about hospital budgets. To avoid such situations, the authorities could consider revising the tripartite structure of the UCM or at least reducing the role of the social partners in the health sector.

Notes

1. In the Luxembourg version of the National Accounts, imputed bank services to non-residents are treated as exports of services and not – as in the 1968 SNA version – as intermediate consumption of resident industries.
2. The interest margin is the difference between the effective interest rates on bank deposits and bank lending.
3. Foreign exchange earnings were 4 per cent of gross earnings in 1997.
4. The acceleration in approved investments also reflected modifications to the diversification law at the beginning of 1997.
5. The majority of cross-border workers comes from France (52 per cent), followed by Belgium (30 per cent) and Germany (18 per cent).
6. The change in definition lowered unemployment by 0.3 percentage point.
7. The European Employment Observatory has calculated its own version of "broad unemployment" (or "extended unemployment") for all EU countries (European Commission, 1998). For Luxembourg, this concept adds to registered unemployment, people participating in active labour market programmes and other programmes to reduce the size of the economic active population (early retirement and disability pensions, parental leave etc.). On this basis, Luxembourg's broad unemployment rate was – at 12.2 per cent in 1996 – still the lowest of all EU countries.
8. The French departments of Meurthe-et-Moselle and Moselle, the German state Saar and the Trier region, and the Belgian provinces of Luxembourg and Liège.
9. Resident Luxembourg nationals earn on average 8.8 per cent more than cross-border workers for similar work and resident foreigners earn 2.6 per cent more (Statec, 1997*a*).
10. The competitiveness indicator shown in Figure 4, Panel A, is equal to the effective exchange rate, deflated by unit labour costs relative to foreign producer prices (for details see Krecké and Pieretti, 1997).
11. Krecké and Pieretti (1997) estimate that the almost 75 per cent of production in the manufacturing sector follows world market prices.
12. The surplus on the current account overstates the national wealth accumulation as – in the absence of reliable statistics – it is not possible to determine the portion of assets that is owned and controlled by Luxembourg citizens.
13. This is also caused by a reclassification of expenditures for railway maintenance following a change in the statutes of the national railway company.
14. In the absence of consolidated government accounts, it is difficult to assess whether general government spending – as a share of GDP – has increased, as part of the increase in central government spending was due to higher transfers to the social security sector.

15. The OECD Forum on Harmful Tax Practices recently established criteria for the identification of preferential tax regimes. Member countries have a 2-year period for self-reviewing their tax regimes.
16. A broad overview of the development of the Luxembourg financial centre can be found in Statec (1995).
17. For further discussion on acceptable versus harmful tax regimes see OECD (1998*a*).
18. Switzerland also decided to abstain from approving the OECD Report or its Recommendations.
19. The rest of the social security sector's receipt are made up by government transfers (45 per cent) and interest receipts (5 per cent).
20. The last time, the institutionalised wage indexation was suspended was between April 1982-September 1984.
21. However, the person may receive a complete disability pension in case of unemployment for a maximum period of 18 months or for an unlimited duration if the person is older than 55.
22. The *préretraite progressive* allows older workers (over 57) to work part-time, if an employer hires a previously unemployed person for the rest of the time.
23. Moreover, only a limited number of unemployed receive an unemployment benefit. In August 1998, only 38 per cent of unemployed received such a benefit.
24. Annually 300 to 400 sanctions are imposed, compared with an annual average of around 2 000 unemployment benefit recipients. Moreover, it has been decided also to apply sanctions to people, who do not receive an unemployment benefit by suspending the treatment of their file for two months, which may delay the start of their benefit.
25. Current legislation on working time, dating back to the early 1970s, provides for normal working time of eight hours a day and 40 hours a week, although collective agreements may specify shorter hours. Deviations are possible under the strictly regulated compensation dispensation scheme and overtime scheme for which Ministerial authorisation may be required.
26. The full social minimum wage applies for workers above the age of 17. Reduced rates for younger workers exist: 17 years, 80 per cent; 16 years, 70 per cent; 15 years, 60 per cent.
27. Evaluations based on employment-subsidy programmes in Australia, Belgium and the Netherlands suggest combined dead-weight and substitution effects as high as 90 per cent.
28. Cash benefits to the sick and those on maternity leave, which are part of the Luxembourg health insurance scheme, are not covered in this chapter.
29. National health spending is estimated at LF 39.2 billion in 1997 (7.1 per cent of GDP). This includes health care expenditure by the resident population including resident officials of Luxembourg-based international institutions. It does not include health care expenditure for cross-border workers. Public health expenditure, including expenditures for cross-border workers, was LF 36 billion in 1997 (6.4 per cent of GDP), of which 80 per cent is spent by the public health insurance (UCM). Expenditures for cross-border workers and other insured residing outside Luxembourg by the UCM was – at only 9 per cent of health insurance spending – well below their share in the insured population (21 per cent, excluding co-insured dependants). In addition, 4 per cent of health insurance spending was for treatment of Luxembourg residents abroad.

30. Luxembourg GDP substantially overestimates the income of the population, given the substantial input of foreign resources in production (cross-border workers and foreign-owned businesses).

31. According to EU rules, non-residents who also receive a pension from their country of residence, are covered by the health system of that country. Therefore, of the 30 000 non-residents who currently receive a Luxembourg pension, only 11 per cent are covered by the Luxembourg health system.

32. Mossialos (1997).

33. The number of acute care beds per 1 000 persons was 5.96 in 1997.

34. An historic overview of the system can be found in OECD (1994).

35. There are in fact nine sickness funds, organised according to sector or profession. These sickness funds only act as intermediaries between the insured and the UCM for the reimbursement of direct payments by the insured members to the providers. The main sickness funds are: the blue-collar workers' sickness fund (*Caisse de maladie des ouvriers*), the private-sector white collar workers' sickness fund (*Caisse de maladie des employés privés*), the civil servants' and public employees' sickness fund (*Caisse de maladie des fonctionnaires et employés communaux*) and the self-employed workers' sickness fund (*Caisse de maladie des professions indépendantes*).

36. International civil servants have their own health insurance and are not in the Luxembourg scheme.

37. The total contribution for the health insurance scheme (including cash benefits for the sick) is 5.4 per cent for white-collar workers and 10.1 per cent for blue-collar workers. This includes a temporary increase of 0.8 percentage point (0.2 percentage point for white-collar workers) to re-stabilise the funds' reserves. It is the intention to lower contributions again in 1999.

38. The *Commission de nomenclature* is a tripartite commission, consisting of representatives of the government, health funds and medical practitioners.

39. The *Conseil supérieur des assurances sociales* consists of a presiding judge and two magistrates, nominated by the Grand Duke for a period of three years.

40. Under current co-payment regulation, the insured are required to pay: 20 per cent for the first medical visit within 28 days and 5 per cent for all other ambulatory medical visits or consultations; 20 per cent for dental care and physiotherapy (maximum of eight visits); 0, 20, 60 or 100 per cent for prescription drugs; Lfr 220 per day at a hospital.

41. In the first case, the Luxembourg health insurance refused to reimburse a pair of glasses that were bought in Belgium, as the person had not asked permission in advance. The Court judged that the Luxembourg rules were an infringement of the free movement of goods and services, as it encouraged people to buy their glasses in Luxembourg (for which no permission is needed). The second ruling involved the treatment of a Luxembourg resident by a Germany based orthodontist. The Luxembourg health insurance did not want to reimburse the person as treatment was also available in Luxembourg. However, the Court judged that orthodontist treatment in the ambulatory sector is a service, and permission in advance would mean an infringement of the free movement of goods and services. The Court has not taken a decision on hospital treatment in another EU country.

42. Chassin *et al.* (1987) found, for the United States, that a sixth to a third of three commonly performed procedures in the fee-for-service system provided zero or even negative clinical benefit. Similar research in France revealed that 20 per cent of some components of medical care is clinically not justified (Béraud, 1992; CNAMTS, 1992).
43. The *Centre hospitalier Luxembourg* is the only hospital where medical practitioners are employed by the hospital.
44. In a capitation payment system, general practitioners receive a fixed payment for each patient on their list, sometimes adjusted for factors such as age and gender. In some countries, the system is combined with some fees or allowances for specific services.
45. In the Diagnostic Related Groups-based system, fees are set according to diagnosed medical conditions and standardised treatment costs.

List of acronyms

ADEM	Public Employment Service
CMCM	*Caisse médico-chirurgicale mutualiste*
CNFPC	*Centre national de la formation professionnelle continue*
DAT	*Division d'auxiliaires temporaires*
EMU	European Economic and Monetary Union
ESCB	European System of Central Banks
EU	European Union
IML	Luxembourg Monetary Institute
Intrastat	Data collection system for intra-EU trade
PPP	Purchasing Power Parity
R&D	Research and Development
RMG	General Assistance Scheme
SMEs	Small and Medium-sized Enterprises
SNA	System of National Accounts
Statec	Luxembourg Bureau for Statistics and Economic Studies
UCM	*Union de caisse de maladie*

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Annex

Calendar of main economic events

1997

January

At the biennial revision, the social minimum wage is increased by 3.2 per cent in line with past wage increases.

For the first time, the 4 000 blue-collar workers in the steel industry sign a separate collective agreement covering only 1996. This change reflects the restructuring of the labour force following the conversion to electric furnace steelmaking.

February

The diversification law is revised, following changes in EU guidelines on state aid for small and medium-sized enterprises and Research and Development projects.

Following the crossing of the threshold value (*cote d'échéance*) of the *échelle mobile*, wages and salaries and all indexed benefits are increased by 2.5 per cent. The previous automatic wage increase was in May 1995.

March

After the blue-collar staff, the 1 800 white-collar workers in the steel industry sign a collective agreement for the period 1996-1997. This includes the adoption of a new profit-sharing scheme.

A new telecommunication bill is adopted in line with EU legislation. One of the objectives of this law is to attract new activities in the area of communications services.

May

A new collective agreement is concluded in the banking sector. In contrast to the previous 1993 agreement, this accord is signed by all the sector's main trade unions. The new collective agreement provides for annual pay rises of 1.7 per cent, 1.6 per cent and 1.5 per cent for the years 1996, 1997 and 1998, respectively, together with a new bonus scheme.

A new electric furnace is put into service in Esch-Belval, which completes the LF 24 billion investment programme in the steel industry.

July

Luxembourg takes over the Presidency of the European Union from the Netherlands. Among its priorities are the ratification of the draft Amsterdam Treaty, pressing ahead with

the final phase of European Economic and Monetary Union and making progress on enlargement of the European Union.

August

The government presents the 1998 budget to Parliament (Chapter II).

The last blast furnace is closed in Esch-Belval, following the starting up of an electric furnace in May.

October

A new collective agreement for blue-collar workers in the steel industry is signed for the years 1997-1998. The overall pay increase comes to 0.92 per cent.

November

The general assembly of the sickness fund (UCM) decides to temporarily raise employers' and employees' contributions for 1998 in order to restabilise the budget (Chapters III and IV).

Luxembourg, as President of the European Union, hosts the Employment Summit, which agrees to a framework for national action under the four "pillars" of employability, adaptability, entrepreneurship and equal opportunities.

December

The seventh satellite of the Astra system is successfully launched from the Cosmodrom in Baikonur (Kazakhstan).

The Public Employment Service (ADEM) and the Union of Temporary Employment Agencies (U*nion luxembourgeoise des entreprises de travail intérimaires*, ULEDI), sign a co-operation agreement. The aim of this deal is to facilitate job-seekers, registered with ADEM, in contacting employers using the services of temporary employment agencies.

1998

January

The Public Employment Service (A*dministration de l'emploi*, ADEM) changes the definition of employment, by removing all participants in labour market programmes from the national employment definition.

White-collar workers in the iron and steel industry sign a new collective agreement for 1998-1999, including an overall pay increase of 1.85 per cent spread over two years and acceptance of a previously agreed profit-sharing scheme.

February

A new collective agreement for the hospital sector is signed, including annual working time reductions from 2 080 to 1 976 hours by the year 2000. The statutory reference period of a month will become more flexible and adapted according to the needs of the enterprise or the wishes of the staff.

April

The Tripartite Co-ordination Committee adopts the National Action Plan for the Promotion of Employment (P*lan d'action national en faveur de l'emploi*) (Chapter III).

The European Court of Justice rules that restrictions by the Luxembourg health system on buying health services from non-domestic providers are against the free movement of goods and services in the European Union (Chapter IV).

May

The Luxembourg airline company Cargolux puts two new aircraft into service.

June

As part of the process towards monetary union, the Luxembourg Monetary Institute becomes the Central Bank of Luxembourg (Chapter II).

July

Parliament adopts the pension reform bill for the public sector, bringing the scheme more in line with the less favourable regime in the private sector (Chapter II).

August

The government presents the 1999 Budget to Parliament.

The eighth satellite of the Astra network is successfully launched from the Cosmodrome of Baikonur in Kazakhstan.

September

The government presents to Parliament a law proposing a new supervisory authority for the financial sector. The *Commission de surveillance du secteur financier* (CSSF) should become operational from January 1999.

November

The stock exchanges of Amsterdam, Brussels and Luxembourg decide to co-operate, by giving their members direct access to each other's markets from January 1999.

BELGIUM

Statistical annex and structural indicators

OECD 1999

Table A. **Belgium – Selected background statistics**[1]

	Average 1988-97	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
A. Per cent changes from previous year											
Private consumption[2]	2.0	3.2	3.7	2.9	2.9	2.2	–1.3	1.6	0.9	1.8	2.1
Gross fixed capital formation[2]	4.0	16.4	11.4	9.6	–4.7	1.3	–3.6	–0.1	4.2	0.5	5.4
Public[2]	0.1	4.6	–26.0	–4.7	8.5	5.1	9.3	6.0	–6.7	–14.9	19.2
Residential[2]	6.0	25.0	17.6	8.2	–9.1	5.0	1.7	5.5	5.5	–4.0	4.9
Business[2]	4.1	15.4	15.3	11.8	–4.2	–0.5	–7.1	–3.4	5.1	4.6	4.2
GDP[2]	2.2	4.7	3.6	3.0	1.6	1.5	–1.5	2.6	2.3	1.3	3.0
Implicit price deflator											
GDP	2.8	2.1	4.6	3.1	3.1	3.7	4.0	2.3	1.5	1.6	1.4
Private consumption	2.6	1.2	4.0	3.3	3.2	2.4	3.2	2.8	2.1	2.3	1.8
Exports of goods and services	1.5	3.7	6.9	–1.5	–0.7	–1.2	–1.2	1.3	0.9	2.5	4.5
Imports of goods and services	1.3	2.2	6.5	–1.5	–0.6	–2.9	–2.6	1.9	2.1	2.9	5.2
Industrial production	1.9	5.8	3.4	3.8	–2.0	0.0	–5.2	1.9	6.5	0.8	4.4
Employment	0.3	1.5	1.6	1.4	0.1	–0.4	–1.1	–1.0	0.5	0.4	0.4
Compensation of employees	4.5	3.5	6.9	7.6	7.9	5.1	3.1	2.7	3.2	1.7	3.1
Productivity (GDP[2]/employment)	1.9	3.2	2.0	1.6	1.4	1.9	–0.4	3.6	1.8	0.9	2.6
Unit labour costs (comp. of employee/GDP[2])	2.2	–1.2	3.2	4.4	6.2	3.6	4.6	0.1	0.8	0.4	0.1
B. Percentage ratios											
Gross fixed capital formation as % of GDP[2]	18.7	17.7	19.0	20.3	19.0	19.0	18.6	18.1	18.4	18.3	18.7
Stockbuilding as % of GDP[2]	0.1	0.4	0.3	0.0	0.1	0.2	0.1	0.3	0.2	0.0	–0.3
Foreign balance as % of GDP[2]	3.1	3.1	2.6	2.2	2.5	2.1	2.2	3.1	4.0	4.0	4.8
Compensation of employee as % of GDP	51.8	51.5	50.8	51.4	52.9	52.9	53.2	52.1	51.8	51.1	50.5
Directs taxes as % of household income	12.8	13.5	12.5	12.8	12.2	12.3	11.9	13.0	13.2	13.1	13.4
Household saving as % of disposable income	17.3	15.3	16.2	16.6	18.2	19.1	20.4	18.4	17.8	16.1	15.2
Unemployment as % of civilian labour force	11.1	11.5	10.4	9.4	8.8	9.4	10.4	12.1	13.1	13.1	12.8
C. Other indicator											
Current balance (BLEU) (billion US$)	8.8	3.5	3.6	3.6	4.9	6.6	11.3	12.5	14.3	14.1	13.7

1. All variables including statistical discrepancies.
2. At constant 1990 prices.

Source: National Accounts Institute and OECD Secretariat.

Table B. **Belgium – Gross domestic product**[1]

Billion francs

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
	Current prices									
Private consumption	3 633.2	3 919.1	4 166.1	4 423.6	4 629.5	4 714.9	4 924.0	5 071.2	5 282.0	5 488.0
Public consumption	857.2	884.0	917.7	987.4	1 027.6	1 088.3	1 139.5	1 181.9	1 205.4	1 251.6
Gross fixed capital formation	1 013.7	1 178.7	1 328.0	1 292.7	1 348.0	1 321.1	1 349.1	1 430.2	1 449.8	1 539.9
Change in stocks	16.1	19.6	–2.9	5.7	7.4	1.4	24.0	32.6	21.5	2.8
Total domestic demand	5 520.2	6 001.4	6 408.9	6 709.3	7 012.5	7 125.7	7 436.6	7 715.9	7 958.7	8 282.4
Exports of goods and services	3 758.1	4 347.6	4 464.4	4 573.0	4 677.8	4 591.7	5 042.6	5 394.6	5 649.4	6 326.4
less: Imports of goods and services	3 588.8	4 181.6	4 318.9	4 413.4	4 461.7	4 312.2	4 710.3	5 042.4	5 303.0	5 933.3
Gross domestic product at market prices	5 689.5	6 167.4	6 554.4	6 868.9	7 228.6	7 405.2	7 768.9	8 068.1	8 305.1	8 675.5
	1990 prices									
Private consumption	3 902.8	4 046.9	4 166.1	4 285.2	4 380.9	4 322.4	4 392.6	4 430.3	4 510.2	4 603.3
Public consumption	931.2	922.3	917.7	937.7	941.0	952.8	968.8	975.2	988.8	997.0
Gross fixed capital formation	1 087.3	1 211.1	1 328.0	1 265.7	1 282.4	1 236.4	1 235.1	1 286.7	1 293.1	1 363.1
Change in stocks	27.4	17.7	–2.9	6.5	12.5	3.3	21.3	16.3	2.4	–20.5
Total domestic demand	5 948.8	6 198.0	6 408.9	6 495.1	6 616.8	6 514.9	6 617.8	6 708.5	6 794.4	6 942.9
Exports of goods and services	3 955.9	4 280.9	4 464.4	4 605.0	4 765.5	4 734.1	5 130.8	5 437.9	5 556.6	5 953.3
less: Imports of goods and services	3 763.6	4 117.5	4 318.9	4 440.9	4 623.7	4 589.3	4 917.6	5 155.7	5 270.1	5 602.7
Gross domestic product at market prices	6 141.1	6 361.4	6 554.4	6 659.2	6 758.6	6 659.7	6 831.0	6 990.7	7 080.9	7 293.5

1. Including statistical discrepancies.

Source: National Accounts Institute.

Table C. **Belgium – Income and expenditure of households and private non-profit institutions**

Billion francs

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Compensation of employees	2 929.5	3 132.9	3 369.7	3 637.0	3 822.7	3 941.5	4 046.2	4 175.7	4 247.2	4 380.0
Income from firms received by individuals	1 147.2	1 181.1	1 262.6	1 294.5	1 359.0	1 390.3	1 517.9	1 570.2	1 645.6	1 704.9
Household property income[1]	741.8	848.7	943.8	1 035.9	1 134.7	1 193.9	1 186.1	1 151.3	1 133.3	1 154.5
Current transfers from government	1 485.3	1 572.3	1 670.7	1 814.1	1 942.0	2 013.1	2 063.0	2 151.4	2 215.8	2 279.3
Current transfers from the rest of the world	46.3	51.4	58.1	59.9	64.0	66.8	67.8	83.2	93.9	111.3
Household income	6 350.1	6 786.4	7 305.0	7 841.4	8 322.4	8 605.6	8 881.0	9 131.8	9 335.9	9 630.1
less:										
Direct taxes	859.7	847.8	938.3	957.1	1 019.8	1 028.3	1 153.7	1 201.7	1 226.9	1 289.4
Social security contribution by wage-earners and self-employed	864.4	916.4	984.6	1 066.1	1 140.3	1 190.1	1 209.9	1 241.5	1 256.7	1 301.4
Current transfers to the rest of the world	54.2	58.6	81.5	80.4	86.8	90.6	96.4	118.7	138.4	139.6
Other current transfers	280.9	284.4	305.5	329.3	350.8	375.5	385.5	403.5	421.6	424.4
Disposable income	4 291.0	4 679.2	4 995.0	5 408.6	5 724.8	5 921.1	6 035.6	6 166.5	6 292.3	6 475.3
Households savings	657.8	760.1	828.9	985.0	1 095.3	1 206.2	1 111.5	1 095.2	1 010.3	987.3

1. Including statistical discrepancy.

Source: National Accounts Institute and OECD, *National Accounts*.

Table D. **Belgium – Income and expenditure of enterprises**

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
	Billion francs									
1. Gross operating surplus excluding subsidies	842	997	994	961	1 013	1 002	1 093	1 216	1 260	1 379
2. Subsidies	176	158	187	203	192	195	188	196	200	176
3. **Gross operating surplus** (1 + 2)	1 018	1 155	1 181	1 164	1 205	1 197	1 280	1 412	1 460	1 555
4. Net property income payable	177	204	258	306	323	284	278	284	285	295
5. **Gross primary income** (3 – 4)	841	951	923	858	883	913	1 003	1 128	1 175	1 260
6. Current transfers pais to other sectors	139	147	126	135	137	158	181	216	221	260
7. **Disposable income** (5 – 6)	702	805	797	722	746	755	822	913	954	1 001
8. Capital transfers (net)	30	21	25	25	32	39	21	–5	31	41
9. **Capital resources** (7 + 8)	731	826	822	748	778	794	843	907	985	1 042
10. Gross capital formation	581	710	798	799	811	755	762	827	858	877
11. Other[1]	17	20	18	20	16	22	29	42	50	42
12. **Net lending**	133	96	6	–71	–48	18	52	39	77	122
	Per cent of GDP									
Gross operating surplus (including subsidies)	17.9	18.7	18.0	16.9	16.7	16.2	16.5	17.5	17.6	17.9
Disposable income	12.3	13.0	12.2	10.5	10.3	10.2	10.6	11.3	11.5	11.5
Capital resources	12.9	13.4	12.5	10.9	10.8	10.7	10.8	11.2	11.9	12.0
Gross capital formation	10.2	11.5	12.2	11.6	11.2	10.2	9.8	10.2	10.3	10.1
Net lending	2.3	1.6	0.1	–1.0	–0.7	0.2	0.7	0.5	0.9	1.4

1. Change in mathematical retirement pension reserves and net purchases of land and intangible assets.

Source: National Accounts Institut and OECD, *National Accounts*.

Table E. **Belgium – Government revenue and expenditure**

Billion francs

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Current revenue	2 227.8	2 289.2	2 464.2	2 630.6	2 715.4	2 832.2	3 085.9	3 284.3	3 431.1	3 647.9
Income froms property and firms	56.9	70.8	78.1	90.2	84.7	85.3	68.5	76.0	89.7	70.1
less: Interest on public debt	–568.2	–629.1	–685.5	–691.6	–771.1	–793.8	–778.5	–719.1	–704.5	–679.4
Indirects taxes	689.4	749.4	801.4	832.8	876.5	918.7	988.7	992.1	1 052.7	1 111.2
Directs taxes	859.7	847.8	938.3	957.1	1 019.8	1 028.3	1 153.7	1 201.7	1 226.9	1 289.4
Social security contributions by wage-earners and self-employed	864.4	916.4	984.6	1 066.1	1 140.3	1 190.1	1 209.9	1 241.5	1 256.7	1 301.4
Imputed social contributions	113.6	119.7	125.5	138.8	146.6	158.8	167.0	173.2	181.2	186.8
Direct taxes on compagnies	157.1	173.8	160.2	165.9	155.5	180.8	213.1	251.6	265.9	305.8
Other current transfers	55.0	40.3	61.5	71.4	63.2	63.9	63.6	67.2	62.6	62.6
Current expenditure	2 483.7	2 577.6	2 725.6	2 961.3	3 102.8	3 236.3	3 346.0	3 474.8	3 580.4	3 684.1
Public consumption	857.2	884.0	917.7	987.4	1 027.6	1 088.3	1 139.5	1 181.9	1 205.4	1 251.6
Subsidies	175.7	158.5	186.6	202.6	192.4	195.4	187.8	196.1	199.7	176.1
Social security transfers to wage-earners and self-employed	1 063.8	1 115.4	1 182.0	1 287.8	1 375.8	1 428.9	1 462.3	1 527.6	1 582.2	1 621.2
Other current transfers (net) to households	136.8	155.5	164.3	170.9	182.8	182.9	197.0	205.6	210.7	221.3
Social benefits corresponding to imputed contributions	119.4	124.8	130.7	143.9	151.6	164.1	171.6	177.7	185.4	190.4
Other current transfers	130.8	139.3	144.2	168.6	172.6	176.8	187.9	185.9	197.0	223.5
Saving of general government	–255.9	–288.4	–261.4	–330.7	–387.4	–404.2	–260.1	–190.5	–149.3	–36.2
Consumption of fixed capital	19.6	21.7	22.2	23.4	24.3	24.9	25.5	26.2	26.5	27.0
Gross saving of general government	–236.4	–266.7	–239.2	–307.3	–363.1	–379.3	–234.6	–164.3	–122.8	–9.1

Source: National Accounts Institute, and OECD, *National Accounts*.

Table F. **Belgium – Area breakdown of foreign trade for the BLEU**

Million $US

	1992	1993[1]	1994	1995	1996	1997
Exports, fob	122 987	117 687	134 301	168 003	168 175	165 104
OECD	108 106	101 875	116 575	145 363	146 136	141 881
EU	96 059	88 434	100 374	126 913	126 492	121 041
of which:						
Germany	28 150	24 650	27 848	36 330	34 439	31 792
France	23 763	22 172	25 288	30 855	30 140	27 936
Netherlands	16 605	15 339	17 563	22 262	22 550	20 625
United Kingdom	9 621	9 850	11 277	13 677	15 356	16 708
Italy	7 250	6 444	6 876	9 300	9 165	9 110
United States	4 773	5 655	6 855	6 721	7 414	8 509
Other OECD	7 275	7 787	9 345	11 729	12 230	12 330
Non-OECD	14 045	15 233	17 082	22 005	21 299	22 571
Ex-COMECOM	1 780	1 272	1 562	884	958	978
OPEC	2 624	2 510	2 146	2 689	2 425	2 670
Other	9 641	11 451	13 374	18 432	17 916	18 924
Unspecified	836	579	645	635	741	652
Imports, cif	124 967	105 427	122 007	153 232	157 916	150 806
OECD	110 354	93 145	106 508	136 108	139 894	131 376
EU	96 045	80 129	91 110	116 087	119 256	108 795
of which:						
Germany	29 872	22 720	24 089	32 202	31 400	27 986
France	20 602	17 090	19 410	23 893	24 216	21 303
Netherlands	21 884	18 430	21 614	27 317	29 323	26 999
United Kingdom	9 624	9 941	11 551	13 566	14 352	13 722
Italy	5 665	4 680	5 184	6 566	6 664	5 885
United States	5 402	5 601	6 468	8 795	9 718	11 628
Other OECD	8 907	7 415	8 930	11 226	10 920	10 953
Non-OECD	14 548	12 161	14 457	16 953	17 952	19 363
Ex-COMECOM	2 087	742	1 206	661	723	900
OPEC	3 309	1 212	1 582	1 415	1 574	1 843
Other	9 153	10 207	11 669	14 876	15 655	16 621
Unspecified	65	121	1 042	172	71	67

1. Following the abolition of customs within the EU on 1 January 1993, data on intra-EU trade are no longer based on customs declarations but on INTRASTAT.

Source: OECD, *Foreign Trade Statistics*, Series A.

Table G. **Belgium – Commodity breakdown of foreign trade for the BLEU**

Million US$

	1990	1993[1]	1994	1995	1996	1997
Exports, fob	118 281	120 631	134 662	168 124	168 375	166 723
SITC sections						
0. Food and live animals	9 852	11 382	12 293	15 138	14 976	14 837
1. Beverages and tobacco	787	1 069	1 206	1 340	1 360	1 313
2. Crude materials, inedible, except fuels	2 692	2 422	2 930	4 030	3 781	3 918
3. Mineral fuels, lubricants and related materials	4 127	4 093	4 072	4 395	5 380	5 038
4. Animal and vegetable oils, fats and waxes	481	403	531	756	696	787
.5 Chemicals and related products, n.e.s.	14 756	16 836	20 679	29 697	28 968	29 941
6. Manufactured goods	36 289	32 694	36 078	46 159	43 508	42 041
7. Machinery and transport equipment	32 287	33 012	37 958	45 950	47 894	47 139
8. Miscellaneous manufactured articles	9 701	10 936	11 296	14 291	15 260	14 782
9. Commodities and transactions, n.e.s.	7 309	7 784	7 620	6 368	6 551	6 926
Imports , cif	120 304	111 037	123 499	153 415	157 967	153 220
SITC sections						
0. Food and live animals	9 426	9 808	10 971	13 780	13 507	12 678
1. Beverages and tobacco	1 305	1 581	1 600	1 913	1 883	1 745
2. Crude materials, inedible, except fuels	6 861	5 034	6 324	7 776	7 058	7 017
3. Mineral fuels, lubricants and related materials	9 800	8 309	8 528	9 431	11 155	10 860
4. Animal and vegetable oils, fats and waxes	405	438	608	727	713	773
5. Chemicals and related products, n.e.s.	13 362	14 213	15 863	21 647	22 196	23 316
6. Manufactured goods	28 095	24 701	27 840	34 687	34 378	33 184
7. Machinery and transport equipment	30 706	27 953	31 552	40 306	44 916	45 606
8. Miscellaneous manufactured articles	12 633	12 665	13 340	15 758	16 634	16 487
9. Commodities and transactions, n.e.s.	7 711	6 335	6 873	7 390	5 527	1 554

1. Following the abolition of customs within the EU on 1 January 1993, data on intra-EU trade are no longer based on customs declarations but on INTRASTAT.

Source: OECD, *Foreign Trade Statitics*, Series C.

Table H. **Belgium – BLEU balance of payments**

Million US$

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Exports										
Goods	87 257	92 008	109 939	107 559	116 869	106 340	122 389	155 048	154 700	150 601
Services	21 041	21 820	28 266	30 511	33 568	33 443	40 289	35 358	35 535	34 891
Income	33 101	48 211	65 322	75 178	88 157	82 961	89 392	72 995	61 114	55 764
Imports										
Goods	84 584	89 734	108 270	105 466	113 186	100 516	115 587	145 500	145 808	142 429
Services	18 938	21 310	26 441	28 766	30 875	30 062	36 314	33 040	32 924	31 947
Income	32 609	45 523	62 970	71 920	85 242	78 049	84 185	66 193	54 068	49 420
Balance										
Goods	2 673	2 274	1 670	2 093	3 683	5 824	6 802	9 548	8 892	8 171
Services	2 102	510	1 825	1 745	2 694	3 381	3 975	2 318	2 611	2 944
Incomes	492	2 688	2 352	3 258	2 915	4 912	5 206	6 802	7 046	6 344
Transfers										
Other transfers, balance	–144	–168	–1 032	–825	–1 123	–970	–1 001	–1 161	–1 312	–908
General government, balance	–1 615	–1 718	–1 188	–1 405	–1 559	–1 815	–2 504	–3 248	–3 126	–2 872
Current balances	3 508	3 586	3 626	4 865	6 610	11 332	12 477	14 259	14 111	13 679
Capital account								377	158	400
Financial account										
Direct investment abroad	–3 766	–6 477	–6 208	–6 291	–11 357	–4 886	–1 204	–11 705	–8 070	–7 953
Portfolio investment, assets	–12 110	–14 391	–9 165	–29 703	–62 337	–58 005	–41 667	–27 793	–47 352	–55 810
Other investment, assets	5 352	–55 262	–64 356	–20 883	–49 300	–52 523	11 363	–22 167	–14 081	–48 189
Direct investment in the BLEU	5 069	7 031	7 953	9 294	11 301	10 776	8 314	10 804	13 784	12 002
Portfolio investment, liabilities	7 772	10 977	7 884	26 992	58 629	49 846	18 221	4 909	36 877	54 988
Other investment, liabilities	–5 077	55 488	62 333	17 201	45 256	41 096	–4 982	33 021	6 461	32 603
Reserve assets (nets)	–827	–312	–494	–515	–616	2 156	–281	–230	–576	–1 016
Errors and omissions (nets)	76	–640	–1 574	–960	1 814	208	–2 238	–1 474	–1 312	–705

Source: OECD Secretariat.

Table I. **Belgium – Structure of output and performance indicators**

I. Structure of output (1990 prices)

	1980	1985	1990	1994	1995	1996	1997	1980	1985	1990	1994	1995	1996	1997
	Share of GDP							Share of total employment						
Agriculture, hunting, forestry and fishing	1.9	2.2	1.9	2.1	2.1	2.0	2.0	3.0	3.0	2.5	2.3	2.3	2.3	
Mining and quarrying	0.3	0.2	0.2	0.2	0.3	0.3	0.3							
Manufacturing	22.3	24.2	23.9	22.9	23.0	22.7	23.1	23.4	21.4	19.9	18.6	18.4	18.0	
of which:														
Food products	4.3	4.4	4.0	4.0	4.0	4.0	4.0	3.0	2.9	2.7	2.7	2.7	2.7	
Textiles	1.5	1.4	1.6	1.5	1.4	1.3	1.3	3.3	2.9	2.5	2.0	1.9	1.7	
Paper	1.1	1.1	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.5	1.4	1.4	1.4	
Chemicals	1.9	3.1	3.3	3.6	3.7	3.8	4.1	2.5	2.6	2.5	2.5	2.4	2.4	
Metals	1.5	1.7	1.7	1.5	1.6	1.5	1.4	2.1	1.7	1.4	1.1	1.1	1.0	
Metals products, machinery and equipment	6.9	7.9	6.9	6.1	6.0	5.9	5.9	9.6	8.4	8.0	7.5	7.5	7.4	
Electricity, gas and water	2.5	2.5	2.5	2.6	2.6	2.7	2.7	1.6	1.5	0.9	0.9	0.8	0.8	
Construction	6.6	4.5	5.2	5.1	5.1	4.9	5.0	8.1	6.1	6.6	7.3	7.1	7.0	
Market services	47.5	48.6	49.2	51.1	51.1	51.4	51.1	42.1	45.0	47.9	49.4	49.8	50.6	
Others[1]	5.8	4.5	5.4	3.9	4.0	4.1	4.3							
Total market sector	81.1	82.3	82.9	84.0	84.1	84.0	84.1	78.3	76.9	77.9	78.5	78.4	78.7	
Non-market sector	13.1	13.2	11.7	12.0	11.9	11.9	11.6	21.7	23.1	22.1	21.5	21.6	21.3	

1. Correction for investment from own resources, intermediate consumption of imputed interests (free services of financial intermediaries), VAT, statistical adjustment and net taxes on imports.

Source: National Accounts Institute; National Bank of Belgium and OECD, *National Accounts*.

Table I. **Belgium – Structure of output and performance indicators** *(cont.)*

2. Economic performance (1990 prices)

	1980	1985	1990	1994	1995	1996	1997	Average 1980-85	1990	1994	1995	1996	1997
	Share of total investment							Productivity growth					
Agriculture, hunting, forestry and fishing	2.0	2.5	1.9	1.1	1.0	1.2	1.2	4.2	–2.0	–4.3	3.3	0.4	
Mining and quarrying	0.4	0.5	0.7	0.3	0.3	0.3	0.3						
Manufacturing	13.1	21.1	27.7	20.6	21.5	23.4	23.4	5.2	1.5	7.6	3.0	2.0	
of which :													
Food products	2.1	2.7	3.4	3.0	2.7	2.9	2.9	2.6	–2.9	0.3	2.7	1.2	
Textiles	0.8	1.8	1.7	1.6	1.2	1.1	1.3	1.8	9.4	5.6	0.6	3.6	
Paper	1.0	1.5	2.1	2.3	2.1	2.0	1.9	2.5	–4.6	11.6	0.4	–1.0	
Chemicals	2.7	4.3	10.2	6.7	7.7	9.4	9.8	10.6	0.1	12.1	8.4	4.4	
Metals	1.6	2.3	1.7	1.6	1.6	1.6	1.5	7.9	1.8	15.5	7.4	–2.1	
Metals products, machinery and equipment	3.6	6.5	6.2	3.5	4.3	4.7	4.2	7.2	1.4	9.5	–0.7	0.8	
Electricity, gas and water	5.1	6.4	3.2	4.6	4.6	5.0	4.9	2.4	14.8	3.7	7.7	7.7	
Construction	1.5	1.7	2.2	1.8	1.9	1.8	1.9	0.0	1.4	0.6	3.8	–1.4	
Market services	57.4	59.7	58.9	61.9	62.3	62.1	62.6	0.6	–0.3	2.7	1.0	–0.2	
Other[1]	4.3	–4.9	–0.2	1.7	1.3	0.2	–1.5						
Total market sector	79.4	92.0	94.7	90.3	91.7	93.8	94.4	2.2	0.4	3.5	1.9	0.4	
Non-market sector	16.2	12.8	5.6	8.0	7.0	6.0	7.2	0.4	0.0	3.9	0.5	2.3	

1. Statistical adjustments and increase in stocks.

Source: National Accounts Institute; National Bank of Belgium and OECD, *National Accounts*.

Table I. **Belgium – Structure of output and performance indicators** *(cont.)*

3. Other indicators (current prices)

	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
R&D as a per cent of GDP in manufacturing sector	4.9	5.0	5.3	5.1	4.8		5.1			4.8	4.8		
Total R&D expenditure as a per cent of total GDP	1.65	1.65	1.65	1.61	1.66		1.63		1.58	1.57	1.59		
Government funded R&D as a per cent of total	31.6	28.7	27.6	26.7	32.0		31.3		32.5	26.9	26.4		
Breakdown of employed workforce by size of establishment:													
1 to 9 employees	14.7	14.9	15.1	16.0	16.1	16.0	16.1	16.3	16.4	16.7	16.7		
10 to 49 employees	20.0	20.4	20.8	21.5	21.8	22.1	22.4	22.5	23.3	23.3	23.3		
50 to 199 employees	20.9	21.0	21.6	21.6	21.4	21.3	21.7	21.8	22.4	22.5	22.7		
200 to 499 employees	14.8	14.4	14.7	14.6	14.9	15.0	14.5	14.8	14.6	14.3	14.2		
500 to 999 employees	9.5	9.6	9.6	9.4	9.4	9.2	9.3	9.3	9.3	9.5	9.3		
1 000 employees or more	20.2	19.8	18.2	17.0	16.5	16.4	15.9	15.4	14.0	13.7	13.8		
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0		
Workforce (thousands)	2 844.6	2 862.5	2 888.7	3 002.6	3 082.0	3 145.3	3 153.5	3 158.9	3 156.2	3 142.0	3 188.0		

Source: National Social Security Office and OECD, *Main Science and Technology Indicators.*

Table J. **Belgium – Labour market indicators**

	1980	1985	1990	1994	1995	1996	1997
A. Trend							
Standardised unemployment rate	8.3	10.3	6.7	10.0	9.9	9.7	9.2
Unemployment rate	7.9	12.3	8.7	12.9	12.9	12.7	12.5
Male	4.6	8.8	5.9	9.8	9.8	9.6	..
Female	13.5	17.7	12.8	17.1	17.0	16.6	..
Youth (15-24 years old)		23.6	14.5	21.8	21.5	20.5	21.3
Share of long-term unemployment[1]	..	68.9	68.7	58.3	62.4	61.3	60.5
Unfilled vacancies (thousands)	5.9	18.3	18.8	19.0	19.7	21.3	24.6
B. Structural and institutional features							
Labour force (per cent change)	0.3	–0.5	0.8	0.2	0.5	0.1	0.2
Participation rate[2]	63.0	61.7	62.6	63.9	64.2	64.2	64.3
Male	78.8	74.2	72.7	72.1	72.1	71.9	71.5
Female	47.0	49.1	52.4	55.5	56.1	56.5	57.0
Employment as a per cent of population aged 15 to 64	58.0	54.3	57.2	55.6	55.9	56.1	..
Employers, self-employed and family and family workers (as a per cent of total employment)	16.4	17.8	17.9	18.9	18.9	19.0	18.8
Employees (as a per cent of total employment)	83.6	82.2	82.1	81.1	81.1	81.0	81.2
Civilian employment by sector (as a per cent of total)							
Agriculture	3.2	3.1	2.7	2.5	2.5	..	..
Industry	34.7	30.2	28.3	27.0	26.5	..	..
Services	62.1	66.7	68.9	70.5	71.0	..	..
of which: General Government	19.2	20.7	20.0	19.1	18.8	..	..
Total	100.0	100.0	100.0	100.0	100.0	..	..
Non-wage labour costs[3]	21.1	24.0	26.8	27.1	27.0	27.0	26.9
Unemployment insurance benefits[4]	33.3	28.8	32.8	..	..	..	..

1. People who have been looking for a job for one year or more as a percentage of total unemployment.
2. Labour force as a percentage of the corresponding population aged between 15 and 64 years.
3. Employers' social security contributions as a percentage of totale wages.
4. Average unemployment benefit as a percentage of compensation per employee.

Source: National Institute of Statistics; *Annuaire statistique de la Belgique*, 1994; Ministry of Finance, *Note de conjoncture* 1996/6/131; OECD, *Labour Force Statistics* (Parts II and III) and *Main Economic Indicators*.

Table K. **Belgium – Public sector**

	1975	1980	1985	1990	1994	1995	1996	1997
Budgetary indicators: general government accounts (as a per cent of GDP)								
Primary receipts[1]	45.0	47.9	51.0	47.3	49.1	49.2	49.4	49.6
Primary total expenditure[1]	47.2	51.3	50.5	43.0	44.6	44.6	44.4	43.9
Primary balance[1]	–2.2	–3.4	0.5	4.2	4.5	4.5	5.0	5.7
Net interest payments	2.9	5.3	9.6	9.7	9.4	8.5	8.1	7.6
Net lending	–5.0	–8.7	–9.1	–5.4	–4.8	–3.9	–3.1	–1.9
Structure of expenditure and taxes (as a per cent of GDP)								
General government expenditure	45.9	52.7	58.0	52.0	53.1	52.0	51.6	50.3
of which:								
Transfers	23.0	25.5	26.8	24.7	26.0	26.0	26.2	26.0
Subsidies	3.1	3.7	3.7	2.8	2.4	2.4	2.4	2.0
Tax receipts	41.7	43.8	46.9	44.0	45.9	45.7	45.8	46.2
Personal income tax	13.5	16.0	16.8	14.3	14.9	14.9	14.8	14.9
Corporate taxes	3.1	2.3	2.6	2.4	2.7	3.1	3.2	3.5
Social Security contributions	13.5	13.2	15.3	15.0	15.6	15.4	15.1	15.0
Consumption taxes	11.7	12.4	12.2	12.2	12.7	12.3	12.7	12.8
of which: Value added tax	6.5	6.8	6.7	6.4	6.4	6.3	6.4	6.5
Other indicators								
Income tax elasticity	1.8	0.6	0.9	1.4	3.8	1.5	0.9	1.6
Income tax as per cent of total tax	39.8	41.6	41.4	38.1	38.3	39.4	39.3	39.8
Gross general government debt (as a per cent of GDP)	58.3	77.1	120.2	125.5	133.3	131.0	126.8	121.9
Net general governemnt debt (as a per cent of GDP)	48.6	68.1	110.1	115.9	124.1	123.8	120.6	116.5

	Prior to		After
Tax rates (per cent)			
Personal income tax rates			
Top rate	72	1 January 1989	55[2]
Lower rate	17	1 January 1989	25[2]
Number of brackets	13	1 January 1989	7
Corporate tax rates	41	1 January 1991	39[2]
Standard VAT rate	20.5	1 January 1991	21.0

1. Excluding interest charges.
2. Excluding the solidarity tax surcharge (*contribution complémentaire de crise*) which has increased the tax rates by 3 per cent since 1 August 1993 (to 56.65,25.75 et 40.17 respectively).

Source: National Bank of Belgium; National Accounts Institute; OECD, *National Accounts* and OCDE Secretariat.

Table L. **Belgium – Financial markets**

1. Sector and Structure of financial assets and liabilities

	1975	1980	1990	1993	1994	1995	1996	1997
Sector size								
Sectoral employment[1]/total employment	..	1.8	2.1	2.0	2.0	2.0		
Non-financial agents' accumulation of financial liabilities/GDP	16.7	21.2	18.7	16.0	10.7	10.1		
Non-financial agents' accumulation of financial assets/GDP	17.5	16.8	18.6	21.7	14.7	14.9		
Stock-market capitalisation/GDP	16.0	9.2	31.6	38.5	34.9	37.6		
Density of banking network[2]	35.9	39.0	36.3	34.8	..	..		
Density of banking network: credit institutions[3]	..	..	90.2	78.1	76.9	75.6		
Structure of financial assets and liabilities								
Share of intermediated financing in total financing[4]	..	..	69.0	67.6	68.1	65.3	63.9	62.2
Financial institutions' share of financial assets of non-financial sectors	..	..	49.8	50.5	50.1	49.6	49.0	48.8
Share of securities issues in financial flows of non-financial agents	..	..	52.1	56.1	57.0	57.7	59.3	60.3
Structure of private non-financial sector's portfolio:[5]								
Deposits	..	..	29.0	26.0	26.9	26.3	25.8	25.1
Bonds and bills	..	..	25.1	23.7	23.9	23.7	22.5	20.6
Equities	..	..	34.7	39.0	38.1	38.2	40.0	42.6
Non-financial corporate financial structure:								
Equity	..	..	34.7	36.5	36.5	37.4		
Short-term debt	..	..	40.3	38.9	58.3	38.2		
Long-term debt	..	..	21.1	20.3	20.9	20.2		

1. Only financial institutions.
2. Number of deposit banks' branches and head offices per 100 000 population.
3. Number of bank branches and head offices per 100 000 population.
4. Share of financial institutions in total external financing of non-financial agents.
5. The private non-financial sector includes corporations as well as households and non-profit institutions. The total differs from 100 because certain items, such as loans and trade credit, are excluded.

Source: National Bank of Belgium.

Table L. **Belgium – Financial markets**

2. Internalisation and efficiency of markets

	1975	1980	1990	1993	1994	1995	1996	1997
Internationalisation of markets								
Foreign business of the banking sector:[1]								
Assets	38.4	46.8	48.6	35.2	35.2	35.2		
Liabilities	43.6	56.1	59.6	40.8	40.2	40.2		
International banking network:								
Foreign banks in Belgium[2]	40	51	63	80	78	78		
Belgian bank branches abroad	. .	. .	52[3]	69	. .	. .		
Share of cross-border transactions:								
Net purchases of foreign securities by residents[4]	25.6	11.1	16.7	76.3	45.3	60.4		
Net purchases of domestic securities by non-residents[5]	9.6	19.4	18.9	49.5	42.5	33.9		
Efficiency of markets								
Cost of bank intermediation[6]	3.1	2.5	1.7	1.9	1.7	1.8		
Bank productivity[7]	73.6	74.6	69.3	67.6	71.6	66.9		
Interest margins[8]	2.5	2.0	1.3	1.2	1.3	1.2		

1. As a percentage of deposit banks' balance sheets.
2. Number of branches and subsidiaries.
3. Number of deposit banks' branches and head offices.
4. Purchases of foreign shares and other securities, equity participations in foreign entreprises as a percentage of total purchases of domestic and foreign securities by the private non-financial sector.
5. Purchases of Belgian shares, foreign equity participations in Belgian entreprises and other purchases of securities issued by residents as a percentage of domestic securities issues.
6. Gross benefit margins as a percentage of the annual average balance sheet of deposit banks.
7. Operating costs as a percentage of the gross benefit margins of deposit banks.
8. Difference between interest receipts and interest payments divided by the annual average balance sheet of deposit banks.

Source: National Bank of Belgium.

LUXEMBOURG

Statistical annex and structural indicators

OECD 1999

Table A. **Luxembourg – Selected background statistics**[1]

	Average 1988-97	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
A. Annual percentage change											
Private consumption[2]	3.2	4.6	5.1	5.7	6.3	–0.9	1.7	2.4	2.4	1.9	2.5
Gross fixed capital formation[2]	7.7	15.0	7.0	2.7	31.6	–9.0	28.4	–14.9	3.5	–1.7	14.1
GDP[2]	5.8	10.4	9.8	2.2	6.1	4.5	8.7	4.2	3.8	3.0	4.8
GDP price deflator	2.5	0.7	4.4	5.2	2.3	2.6	0.6	4.7	0.3	2.2	2.5
Industrial product	2.4	8.7	7.8	–0.5	0.3	–0.8	–4.3	5.9	1.4	–1.9	7.2
Employment	3.0	3.0	3.5	4.1	4.1	2.5	1.7	2.5	2.6	2.7	3.2
Compensation of employees	7.8	7.1	12.2	11.2	9.9	9.3	5.7	7.4	4.2	5.2	5.4
Productivity (GDP[2]/employment)	2.7	7.2	6.1	–1.9	2.0	1.9	6.9	1.6	1.1	0.4	1.6
Unit labour costs (compensation of employees/GDP)[2]	2.3	–3.0	2.1	8.2	4.9	3.6	–1.5	2.7	1.2		
B. Percentage ratios											
Gross fixed capital formation as % of GDP[2]	25.3	24.0	23.4	23.5	29.2	25.4	30.0	24.5	24.5	23.3	25.4
Stockbuilding as % of GDP[2]	0.8	2.9	–0.4	1.2	4.6	2.8	2.1	–0.6	–1.1	–1.8	–1.7
Foreign balance as % of GDP[2]	8.5	4.8	6.0	5.1	2.9	8.2	7.8	11.6	12.2	13.2	13.2
Compensation of employees as % of GDP	54.9	54.2	53.1	54.6	56.0	56.5	55.4	54.3	54.8		
Number of unemployed	3 718	2 483	2 269	2 060	2 298	2 734	3 526	4 643	5 130	5 680	6 354
Unemployment as % of civilian labour force	2.2	1.6	1.4	1.3	1.4	1.6	2.1	2.7	3.0	3.3	3.6
C. Other indicator											
Current balance (million US$)	2 022	1 221	1 602	1 753	1 475	1 944	1 760	2 714	2 875	2 809	2 072

1. Troughout this *Survey*, GDP is on a SNA basis.
2. 1990 prices.

Source: STATEC, preliminary constant data price and *Note de conjoncture* No. 3/98; OECD, *National Accounts* and OECD Secretariat.

Table B. **Luxembourg – Gross domestic product**

	1988	1989	1990	1991	1992	1993	1994	1995	1996[1]	1997[1]
	Current prices									
Private consumption	179.2	192.0	206.3	225.6	229.2	239.0	253.7	262.6	270.8	281.7
Public consumption	37.7	41.6	45.9	49.1	52.6	56.7	60.3	66.5	71.3	72.8
Gross fixed capital formation	68.5	73.1	84.5	103.1	98.5	114.7	107.2	116.9	115.1	134.5
Change in stocks	2.4	1.6	4.1	3.3	1.7	0.5	1.9	4.7	2.6	3.8
Total domestic demand	287.8	308.4	340.8	381.1	382.1	410.9	423.1	450.6	459.8	492.8
Exports of goods and services	295.2	347.4	365.8	392.0	407.6	453.5	501.7	514.4	539.5	594.2
less: Imports of goods and services	291.6	321.7	347.6	383.2	371.9	407.6	426.2	446.2	453.0	500.0
Gross domestic product at market prices	291.5	334.1	359.0	389.9	417.8	456.8	498.6	518.8	546.3	587.0
	1990 prices									
Private consumption	185.7	195.2	206.3	219.3	217.3	220.9	226.3	231.7	236.1	242.0
Public consumption	42.8	44.5	45.9	47.7	48.4	50.2	51.2	52.3	54.1	56.2
Gross fixed capital formation	76.9	82.3	84.5	111.2	101.2	129.9	110.6	114.5	112.5	128.4
Change in stocks	9.2	–1.4	4.1	17.5	11.1	9.0	–2.7	–5.1	–8.6	–8.4
Total domestic demand	314.6	320.5	340.8	395.7	378.0	410.0	385.4	393.4	394.1	418.1
Exports of goods and services	327.4	353.8	365.8	390.1	408.7	420.1	438.4	457.6	468.2	496.2
less: Imports of goods and services	311.9	332.6	347.6	378.9	375.9	386.5	385.9	400.6	404.7	429.3
Gross domestic product at market prices	319.9	351.4	359.0	381.1	398.1	432.8	451.0	468.0	482.3	505.4

1. Estimates.

Source: STATEC, *Note de conjoncture* No. 3/98 and preliminary data; OECD, *National Accounts* and OECD Secretariat.

Table C. **Luxembourg – Labour market**

Thousands

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997[1]
Labour force[2] **(A + B)**	176.1	182.0	189.2	197.1	202.4	206.7	212.9	218.9	225.3	232.8
A. Unemployed	2.5	2.3	2.1	2.3	2.7	3.5	4.6	5.1	5.7	6.4
B. Total employment	173.6	179.7	187.1	194.8	199.7	203.2	208.3	213.8	219.6	226.5
Employees	156.3	162.7	170.4	178.4	183.4	187.0	192.2	197.5	203.1	209.9
Agriculture	1.3	1.4	1.5	1.5	1.5	1.5	1.6	1.6	1.6	1.6
Industry	35.6	36.2	36.0	36.2	35.7	34.1	33.1	32.5	32.2	32.0
of which: Iron and steel	13.5	13.1	12.3	11.6	10.9	10.0	9.4	8.9	8.3	7.7
Energy	1.5	1.5	1.5	1.6	1.6	1.6	1.6	1.7	1.8	1.8
Construction	15.9	16.8	18.3	20.2	21.6	22.1	21.9	22.1	22.4	22.6
Market services	76.0	80.5	85.9	90.8	94.1	97.8	102.8	107.4	111.9	117.6
Non-market services	26.0	26.4	27.1	28.1	28.9	29.8	31.2	32.2	33.2	34.4
Self-employed and family helpers	17.3	17.0	16.7	16.4	16.3	16.2	16.1	16.3	16.4	16.6
Unemployment rate (per cent)	1.6	1.4	1.3	1.4	1.6	2.1	2.7	3.0	3.3	3.6
Participation rate (per cent)	61.2	61.8	62.1	62.1	61.7	61.3	61.8	61.7	62.1	62.5
Job vacancies[3]										
Unfilled vacancies (monthly average)	216.3	201.0	177.7	153.8	159.0	125.6	271.3	437.7	1 107.3	931.4

1. Estimates.
2. Including cross-border workers, nets.
3. In units.

Source: STATEC; OECD, *Main Economic Indicators* and OECD Secretariat.

Table D. **Luxembourg – Structure of output and performance indicators**

1. Structure of output and economic performance (1990 prices)

	1985	1990	1994	1995	1996[1]	1997[1]	1985	1990	1994	1995	1996[1]	1997[1]
	Share of GDP						Share of total employees					
Structure of output												
Agriculture, forestry and fishing	2.0	1.6	0.9	0.9	0.8	0.8	0.9	0.9	0.8	0.8	0.8	0.7
Energy and water	1.9	1.6	1.5	1.5	1.5	1.4	1.0	0.9	0.8	0.8	0.9	0.9
Mining, quarrying and manufacturing	23.1	21.3	15.1	15.1	14.5	14.3	25.8	21.1	17.2	16.4	15.9	15.2
of which: Ores and metals	10.1	6.4	2.8	2.6	2.6	2.9	10.8	7.2	4.9	4.5	4.1	3.6
Construction	4.5	7.0	6.6	6.5	5.8	5.5	9.2	10.8	11.4	11.2	11.0	10.8
Market services[2]	58.0	56.3	64.0	64.0	65.1	65.8	46.1	50.4	53.5	54.4	55.1	56.0
of which: Financial institutions and insurance companies	9.8	13.3	15.0	15.0	15.3	15.3	7.7	9.9	9.5	9.5	9.4	9.2
Non-market services[3]	10.5	12.1	11.9	12.1	12.2	12.2	17.0	15.9	16.2	16.3	16.4	16.4
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	Share of total investment						Productivity growth[4]					
Economic performance												
Agriculture, forestry and fishing							2.8	0.9	–1.3	8.8		
Energy and water							3.4	1.6	2.1	3.1		
Mining, quarrying and manufacturing							7.0	–1.6	0.6	4.9		
of which: Ores and metals							6.4	2.2	–9.5	1.6		
Construction							1.2	–2.7	–0.6	2.0		
Market services[2]							5.0	1.7	0.9	0.5		
of which: Financial institutions and insurance companies							4.5	–2.5	0.6	3.5		
Non-market services[3]							0.7	0.8	–0.6	–2.6		

1. Estimates.
2. Wholesale and retail trade; restaurant and hotels; transport, storage and communications; finance, insurance, real estate and business services.
3. Community, social and personal services.
4. Average for 1985-89 instead of 1985.

Source: STATEC, *Note de conjoncture* No. 3/98 and preliminary constant price data and OECD, *National Accounts* and OECD Secretariat.

Table D. **Luxembourg – Structure of output and performance indicators** *(cont.)*

2. Other indicators

	1980	1985	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Breakdown of employed workforce by size od establishment: [1]												
1 to 19 employees	13.6	14.4	13.9	13.9	13.7	15.0	14.9	16.1	16.8	16.7		
20 to 49 employees	9.2	10.2	11.5	11.9	12.7	13.0	14.4	14.2	14.6	15.3		
50 to 99 employees	8.5	9.2	10.4	11.7	11.2	10.1	10.4	12.0	11.4	10.5		
100 to 249 employees	14.3	16.0	14.9	14.8	15.0	17.1	15.8	17.4	17.2	18.3		
250 to 499 employees	12.0	11.9	9.6	8.7	10.1	10.8	12.5	11.0	10.9	12.2		
500 employees and over	42.3	38.3	39.7	39.0	37.3	34.0	32.0	29.3	29.0	27.0		
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0		
Workforce (thousands)	57.4	52.0	55.0	56.0	56.5	59.8	60.3	59.8	58.4	59.2		

1. Only industry and construction.

Source: STATEC.

Table E. **Luxembourg – Public sector**

	1985	1990	1993	1994	1995	1996	1997
Budget indicators: general government accounts (as a per cent of GDP)							
Primary receipts [1]	49.6						
Primary expenditure [1]	46.0						
Primary balance [1]	3.6						
Nets interest payments	–1.2						
General government budget balance	4.8	5.0	1.7	2.8	1.9	2.9	3.0
Structure of expenditure and taxes (as a per cent of GDP)							
General government current expenditure	47.1						
of which:							
Subsidies	3.5	3.2	3.0	3.0	2.2		
Social security benefits	24.4	24.0	25.5	25.0	25.5	26.1	
Investment	4.4	4.6	5.4	4.4	4.7	4.9	
Tax receipts	46.7	43.4	43.9	44.3	44.1	44.4	
Personal income tax	12.0	10.2	9.1	9.5	9.4	9.8	
Corporate tax	8.3	6.9	7.1	7.5	7.7	7.2	
Social security contributions	12.3	11.8	12.3	11.8	11.8	11.6	
Consumption taxes	11.4	10.8	12.0	12.1	12.0	12.4	
of which: Value added tax	5.8	5.9	6.3	6.0	6.2	6.7	
Other indicator							
Income tax as a per cent of total tax	43.3	39.4	37.0	38.5	38.8	38.3	

	Prior to		After
Tax rates (per cent)			
Personal income tax rates			
Top rate	56	6 December 1990	50
Lowest rate	10	6 December 1990	10
Number of brackets	25	6 December 1990	18
Corporate tax rates	34	6 December 1990	33
VAT rates			
Lower reduced rate		1 January 1992	3
Reduced rate		1 January 1992	6
"Parking" rate		1 January 1992	12
Standard rate	12	1 January 1992	15

1. Excluding interest charges.

Source: STATEC, *Rapport annuel 2/98* and *Annuaire statistique 1997;* Ministry of Finance and OECD, *Revenue Statistics* and *National Accounts.*

Table F. **Luxembourg – Financial markets**

	1980	1985	1990	1991	1992	1993	1994	1995	1996	1997
Sector size										
Sector employment [1]/total employment	4.8	6.3	8.7	8.8	8.8	9.1	9.5	9.4		
Financial assets/GDP [2]	26.7	33.7	36.3	34.4	36.2	36.4	36.8	35.8		
Stock-market capitalisation/GDP	30.9	59.5	99.5	99.0	94.4	177.1	220.3	216.3		
Density of banking network [3]	30.5	32.2	46.7	48.6	54.6	55.2	55.4	53.7		
Structure of financial flows										
Share of credits granted to non-financial sector in total banking assets	35.1	33.4	24.0	24.4	24.7	23.8	18.4	18.1		
Internationalisation of markets										
Foreign business of the banking sector: [4]										
Assets	84.2	86.1	88.1	87.9	86.6	81.5	81.9	80.8		
Liabilities	78.4	77.6	76.5	75.9	73.7	73.4	73.0	72.4		
International banking network:										
Foreign banks in Luxembourg [5]	100	107	158	169	205	210	214	212		
Luxembourg bank branches abroad [6]	74	70	55	6	6	7	7	11		
Efficiency of markets										
Interest margins [7]	0.8	1.2	0.8	0.8	0.8	0.7	0.7	0.7		
Bank productivity [8]	40.5	27.9	37.3	40.5	39.4	38.0	44.8	46.3		
Cost of bank intermediation [9]	0.9	1.4	1.2	1.2	1.1	1.2	1.1	1.1		

1. Credit institutions until 1990, credit institutions plus other intermediaries from 1991.
2. Ratio of banks' balance sheet total to GDP.
3. Number of saving and banking institutions per 100 000 population.
4. As a percentage of deposit banks' balance sheets.
5. Number of foreign saving and banking institutions.
6. Number of regional offices, agencies and branches abroad.
7. Interest margins divided by total assets.
8. Operating expenses as a per cent of gross earnings.
9. Gross earnings as a per cent of total assets.

Source: OECD, *Bank Profitability* 1985-1994; STATEC, *Rapport annuel 1995* and data provided by the Institut monétaire Luxembourgeois.